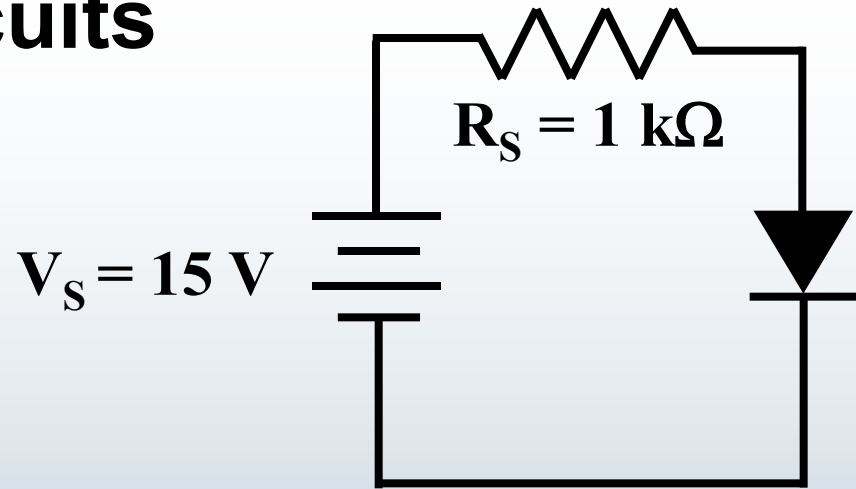


EEE 111

Diode Applications

Topic 2 (Chapter 2)

Solving Diode Circuits



A circuit like this can be solved in several ways:

1. Use the first approximation or ideal equivalent circuit

$$V_D = V_{FB} = V_{ON} = 0\text{V}; I_D = I_{RB} = I_{OFF} = 0\text{A}$$

2. Use the second approximation or simplified equivalent circuit or constant voltage drop model

$$V_D = V_{FB} = V_{ON} = 0.7\text{V}(\text{Si})/0.3\text{V}(\text{Ge})/1.2\text{V}(\text{GaAs})$$

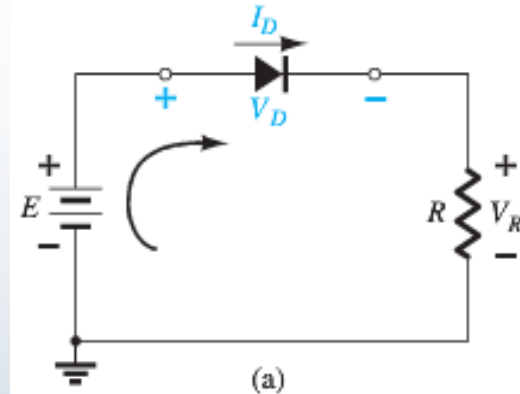
$$I_D = I_{RB} = I_{OFF} = 0\text{A}$$

3. Use the diode's characteristic curve (load line analysis)

2.2 Load-Line Analysis

$$+E - V_D - V_R = 0$$

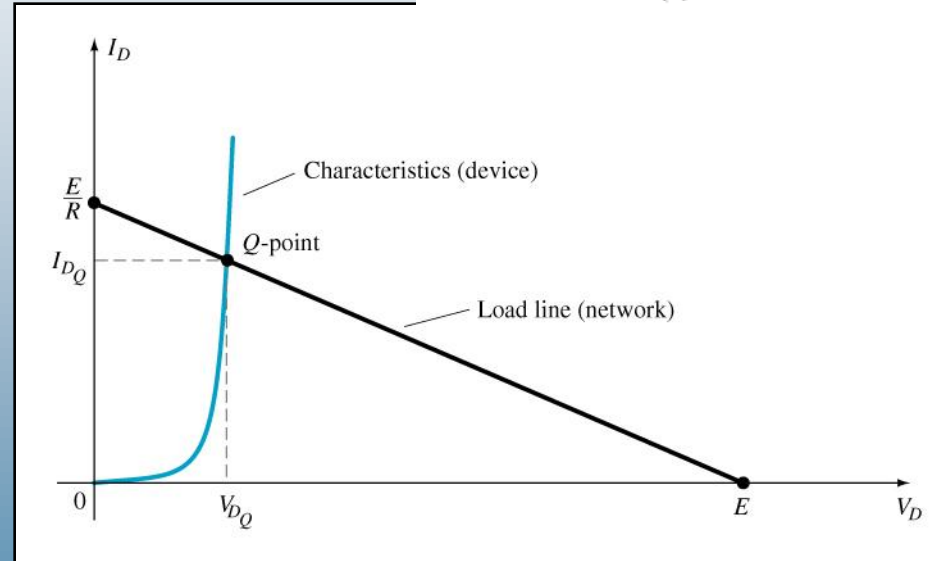
$$E = V_D + I_D R$$



$$V_D = 0V \rightarrow I_D = \frac{E}{R} = I_{MAX}$$

$$I_D = 0A \rightarrow V_D = E = V_{MAX}$$

The load line plots all possible combinations of diode current (I_D) and voltage (V_D) for a given circuit. The maximum I_D equals E/R , and the maximum V_D equals E .



The point where the load line and the characteristic curve intersect is the Q-point, which identifies I_D and V_D for a particular diode in a given circuit.

Load-Line Analysis

- The solution found by load-line analysis is similar to that obtained by a simultaneous solution of

$$I_D = \frac{E}{R} - \frac{V_D}{R}$$
$$I_D = I_s(e^{V_D/nV_T} - 1)$$

(assuming this theoretical diode curve is very close to the actual curve.)

- The mathematics involved would require the use of nonlinear techniques that are beyond the needs and scope of this course.

EXAMPLE 2.1

- For the series diode configuration of Fig. 2.3a, employing the diode characteristics of Fig. 2.3b, determine:

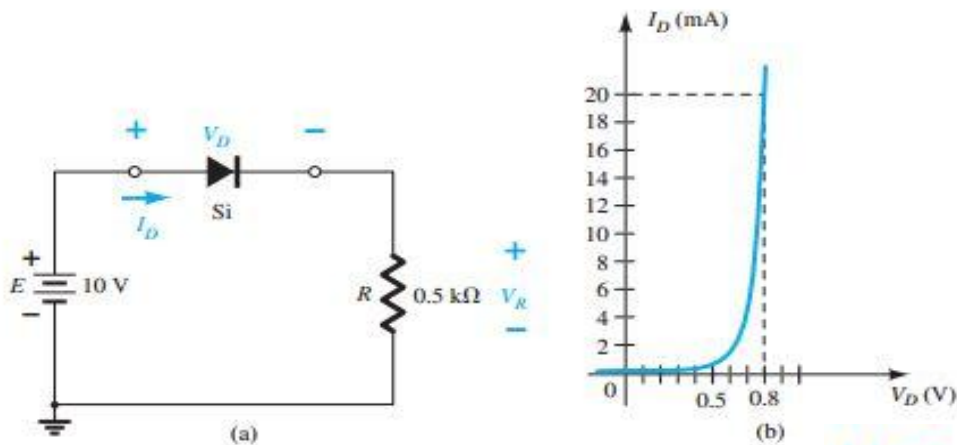


FIG. 2.3

(a) Circuit; (b) characteristics.

Solution:

a. Eq. (2.2):
$$I_D = \frac{E}{R} \Big|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$$

Eq. (2.3):
$$V_D = E \Big|_{I_D=0\text{ A}} = 10\text{ V}$$

The resulting load line appears in Fig. 2.4. The intersection between the load line and the characteristic curve defines the *Q*-point as

$$V_{DQ} \cong 0.78\text{ V}$$

$$I_{DQ} \cong 18.5\text{ mA}$$

The level of V_D is certainly an estimate, and the accuracy of I_D is limited by the chosen scale. A higher degree of accuracy would require a plot that would be much larger and perhaps unwieldy.

b. $V_R = E - V_D = 10\text{ V} - 0.78\text{ V} = 9.22\text{ V}$

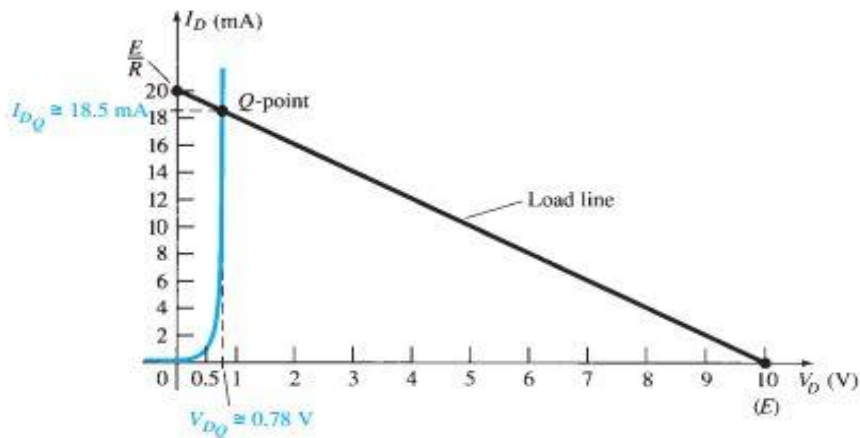
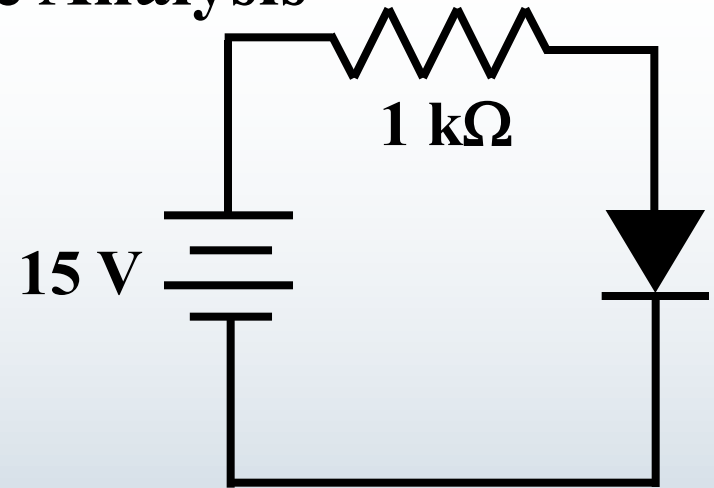
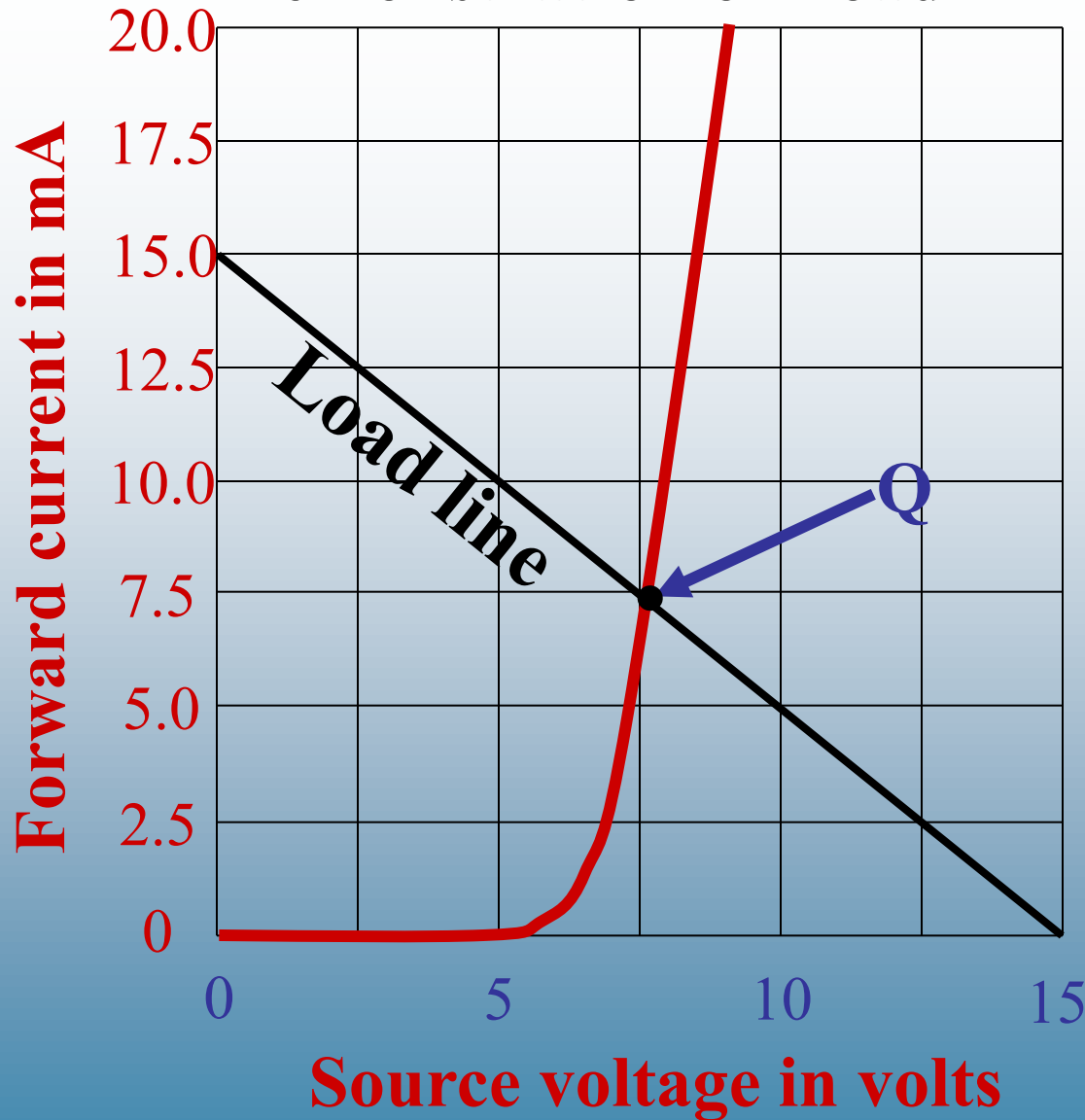


FIG. 2.4

Solution to Example 2.1.

Demonstration of Load Line Analysis

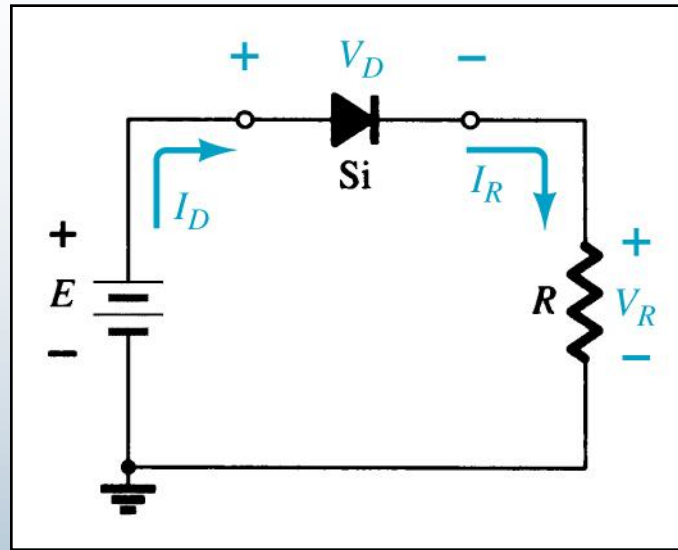


$$I_{\text{SAT}} = \frac{15 \text{ V}}{1 \text{ k}\Omega} = 15 \text{ mA}$$

$$V_{\text{CUTOFF}} = 15 \text{ V}$$

Q is the operating point or quiescent point

2.3 Series Diode Configurations



- The forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored.
- In general, a diode is in the “on” state if the current established by the applied sources is such that its direction matches that of the arrow in the diode symbol, and $V_D \geq 0.7 \text{ V}$ for silicon, $V_D \geq 0.3 \text{ V}$ for germanium, and $V_D \geq 1.2 \text{ V}$ for gallium arsenide.

2.3 Series Diode Configurations

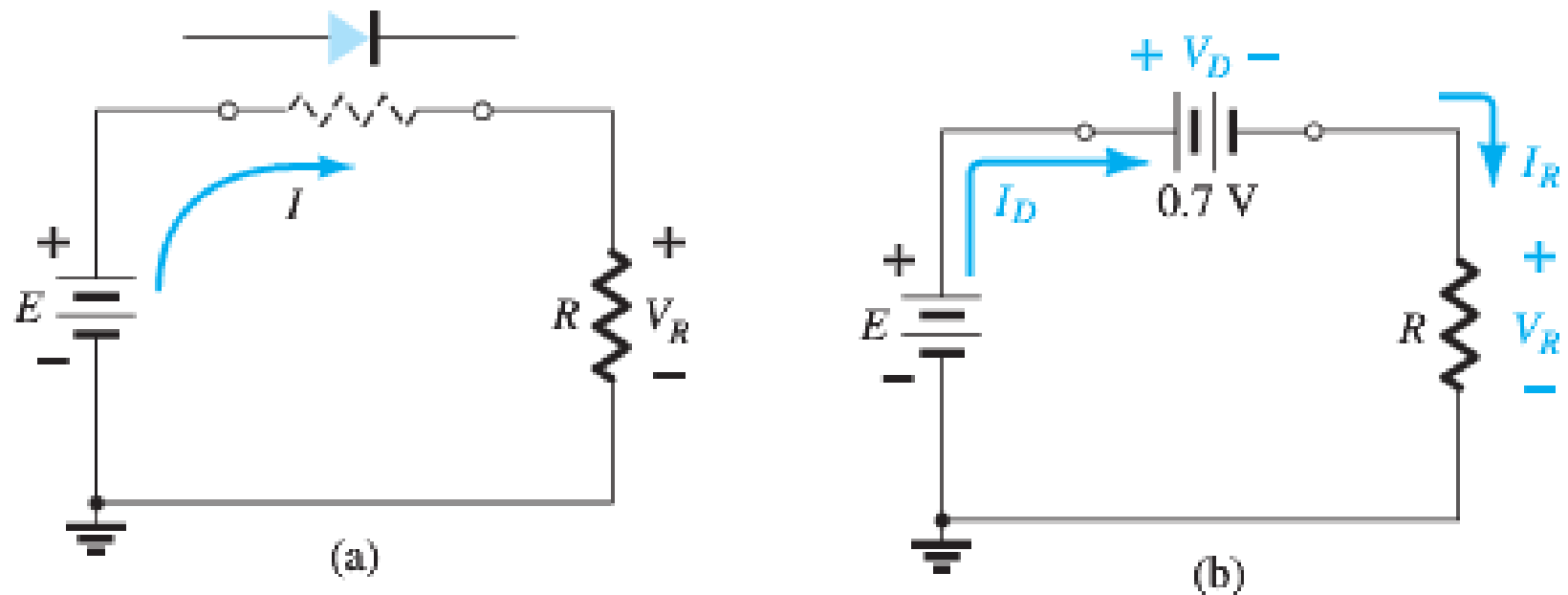


FIG. 2.9

(a) Determining the state of the diode of Fig. 2.8; (b) substituting the equivalent model for the “on” diode of Fig. 2.9a.

$$V_D = V_K$$

$$V_R = E - V_K$$

$$I_D = I_R = \frac{V_R}{R}$$

Reversing the diode

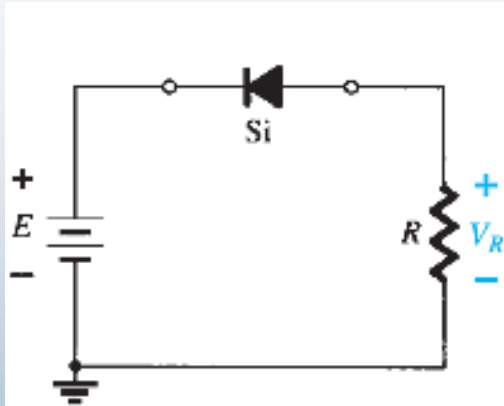


FIG. 2.10

Reversing the diode of Fig. 2.8.

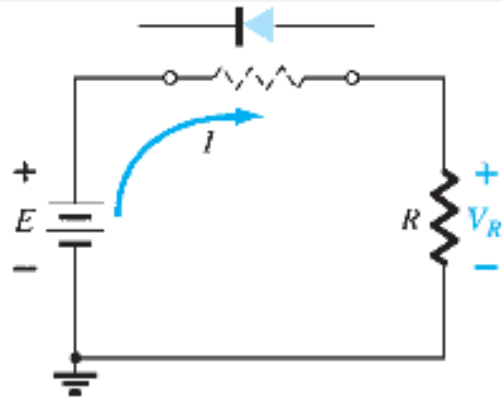


FIG. 2.11

Determining the state of the diode of Fig. 2.10.

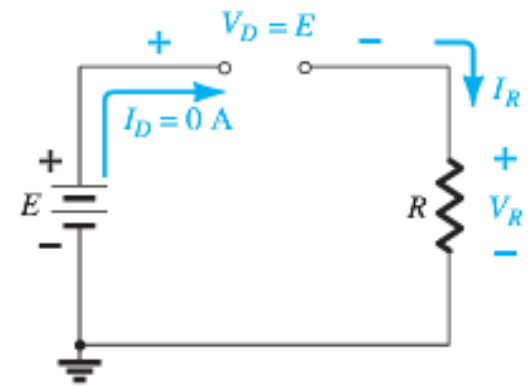


FIG. 2.12

Substituting the equivalent model for the “off” diode of Fig. 2.10.

$$V_R = I_R R = I_D R = (0 \text{ A})R = 0 \text{ V}$$

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D , V_R , and I_D .

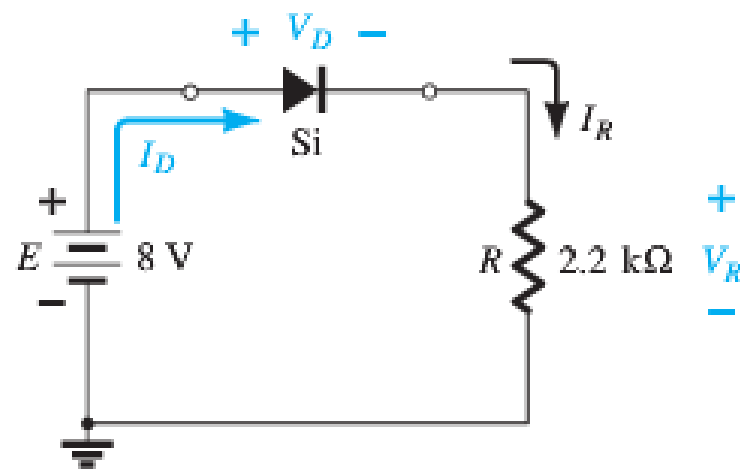


FIG. 2.13

Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = 0.7\text{ V}$$

$$V_R = E - V_D = 8\text{ V} - 0.7\text{ V} = 7.3\text{ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong 3.32\text{ mA}$$

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = 0$ A due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = 0$ V. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8 \text{ V}$$

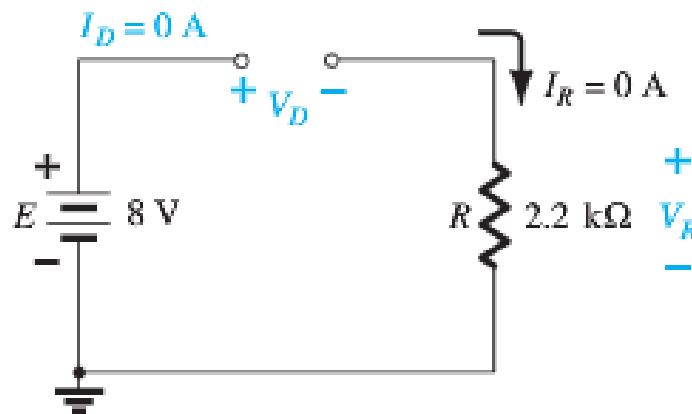


FIG. 2.14

Determining the unknown quantities for Example 2.5.

EXAMPLE 2.6 For the series diode configuration of Fig. 2.16, determine V_D , V_R , and I_D .

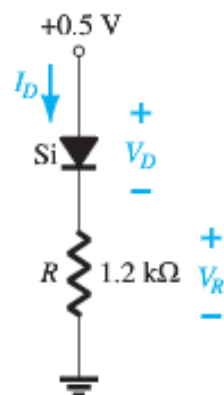


FIG. 2.16

Series diode circuit for Example 2.6.

Solution: Although the “pressure” establishes a current with the same direction as the arrow symbol, the level of applied voltage is insufficient to turn the silicon diode “on.” The point of operation on the characteristics is shown in Fig. 2.17, establishing the open-circuit equivalent as the appropriate approximation, as shown in Fig. 2.18. The resulting voltage and current levels are therefore the following:

$$I_D = 0 \text{ A}$$

$$V_R = I_R R = I_D R = (0 \text{ A}) 1.2 \text{ k}\Omega = 0 \text{ V}$$

$$V_D = E = 0.5 \text{ V}$$

and

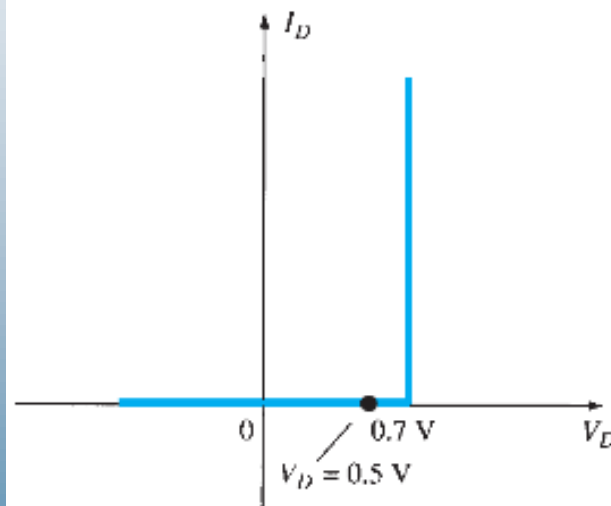


FIG. 2.17

Operating point with $E = 0.5 \text{ V}$.

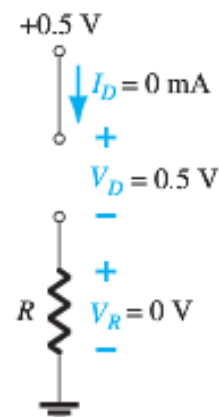


FIG. 2.18

Determining I_D , V_R , and V_D for the circuit of Fig. 2.16.

EXAMPLE 2.7 Determine V_o and I_D for the series circuit of Fig. 2.19.

Assume 1.8V forward voltage drop for the red LED.

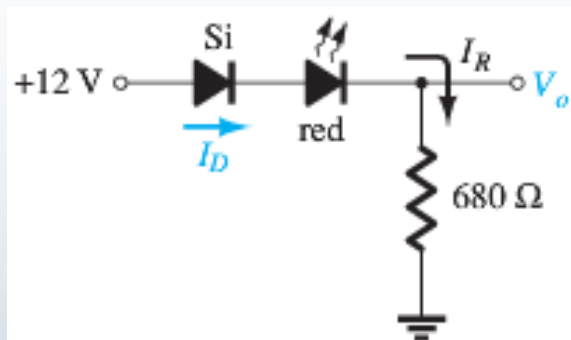


FIG. 2.19

Circuit for Example 2.7.

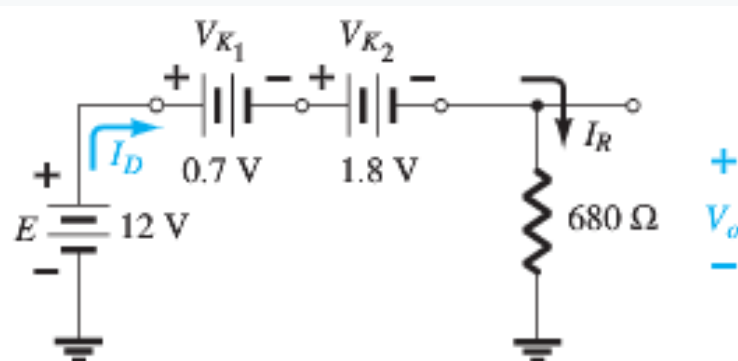


FIG. 2.20

Determining the unknown quantities for Example 2.7.

Solution: An attack similar to that applied in Example 2.4 will reveal that the resulting current has the same direction as the arrowheads of the symbols of both diodes, and the network of Fig. 2.20 results because $E = 12\text{ V} > (0.7\text{ V} + 1.8\text{ V [Table 1.8]}) = 2.5\text{ V}$. Note the redrawn supply of 12 V and the polarity of V_o across the 680-Ω resistor. The resulting voltage is

$$V_o = E - V_{K1} - V_{K2} = 12\text{ V} - 2.5\text{ V} = \mathbf{9.5\text{ V}}$$

and

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{9.5\text{ V}}{680\ \Omega} = \mathbf{13.97\text{ mA}}$$

EXAMPLE 2.8 Determine I_D , V_{D_2} , and V_o for the circuit of Fig. 2.21.

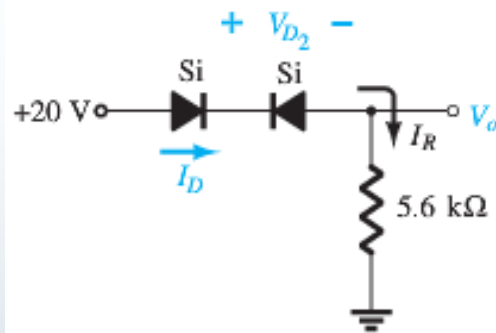


FIG. 2.21

Circuit for Example 2.8.

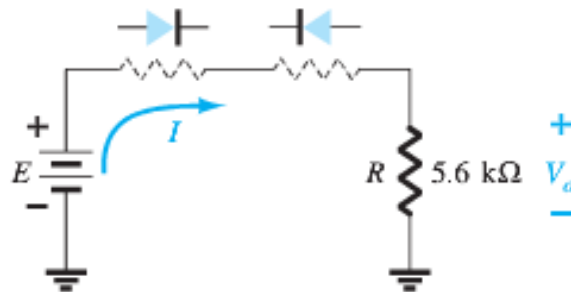


FIG. 2.22

Determining the state of the diodes of Fig. 2.21.

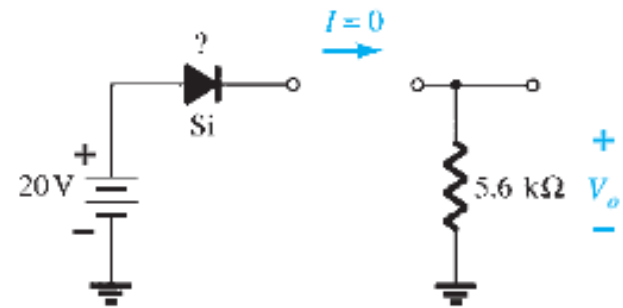


FIG. 2.23

Substituting the equivalent state for the open diode.

For an actual practical diode, when $I_D = 0$ A, $V_D = 0$ V (and vice versa) **No Bias condition**

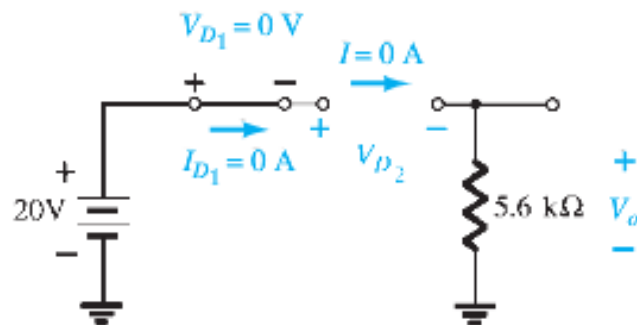


FIG. 2.24

Determining the unknown quantities for the circuit of Example 2.8.

$$V_o = I_R R = I_D R = (0 \text{ A}) R = 0 \text{ V}$$

and

$$V_{D_2} = V_{\text{open circuit}} = E = 20 \text{ V}$$

Applying Kirchhoff's voltage law in a clockwise direction gives

$$E - V_{D_1} - V_{D_2} - V_o = 0$$

and

$$V_{D_2} = E - V_{D_1} - V_o = 20 \text{ V} - 0 - 0 = 20 \text{ V}$$

with

$$V_o = 0 \text{ V}$$

EXAMPLE 2.9 Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig. 2.25.

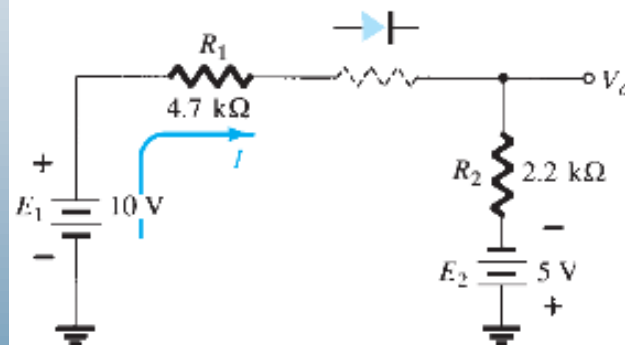
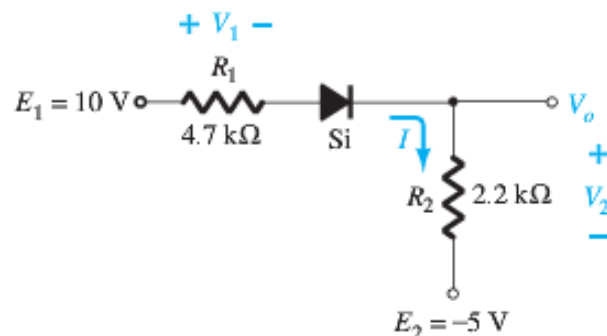


FIG. 2.26

Determining the state of the diode for the network of Fig. 2.25.

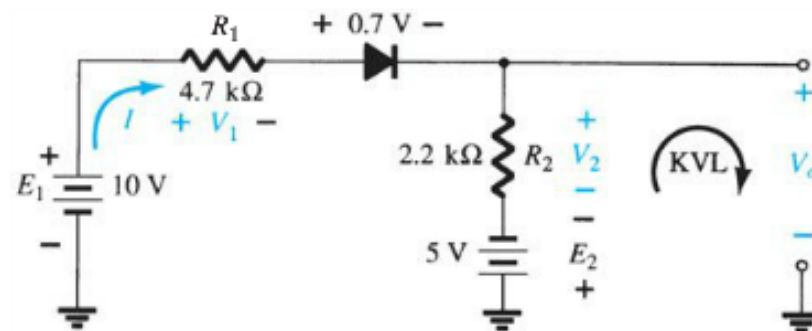


FIG. 2.27

Determining the unknown quantities for the network of Fig. 2.25. KVL, Kirchhoff voltage loop.

The resulting current through the circuit is

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega} \cong 2.07\text{ mA}$$

$$V_1 = IR_1 = (2.07\text{ mA})(4.7\text{ k}\Omega) = 9.73\text{ V}$$

$$V_2 = IR_2 = (2.07\text{ mA})(2.2\text{ k}\Omega) = 4.55\text{ V}$$

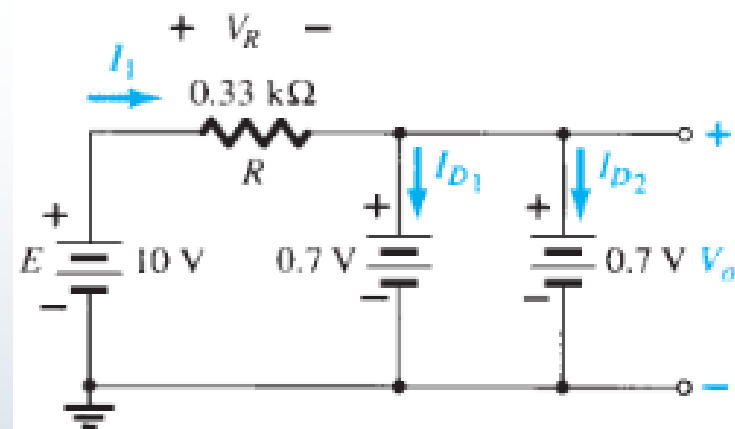
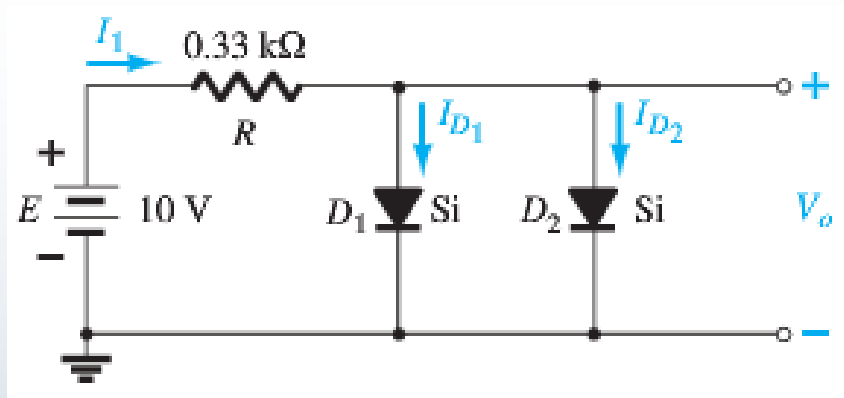
$$-E_2 + V_2 - V_o = 0$$

$$V_o = V_2 - E_2 = 4.55\text{ V} - 5\text{ V} = -0.45\text{ V}$$

2.4 PARALLEL AND SERIES–PARALLEL CONFIGURATIONS

- The methods applied for series diode configurations can be extended to the analysis of parallel and series–parallel configurations.
- For each area of application, simply match the sequential series of steps applied to series diode configurations.

EXAMPLE 2.10 Determine V_o , I_1 , I_{D_1} , and I_{D_2} for the parallel diode configuration of Fig. 2.28.



$$V_o = 0.7 \text{ V}$$

The current is

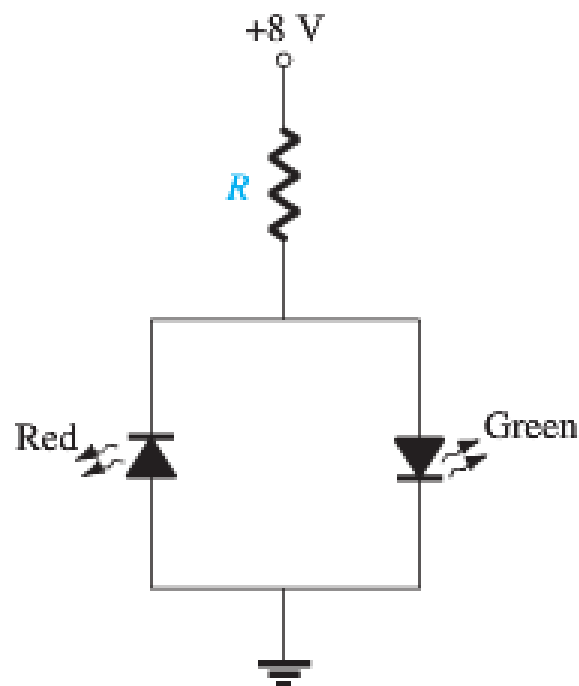
$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{0.33 \text{ k}\Omega} = 28.18 \text{ mA}$$

Assuming diodes of similar characteristics, we have

$$I_{D_1} = I_{D_2} = \frac{I_1}{2} = \frac{28.18 \text{ mA}}{2} = 14.09 \text{ mA}$$

EXAMPLE 2.11 In this example there are two LEDs that can be used as a polarity detector. Apply a positive source voltage and a green light results. Negative supplies result in a red light. Packages of such combinations are commercially available.

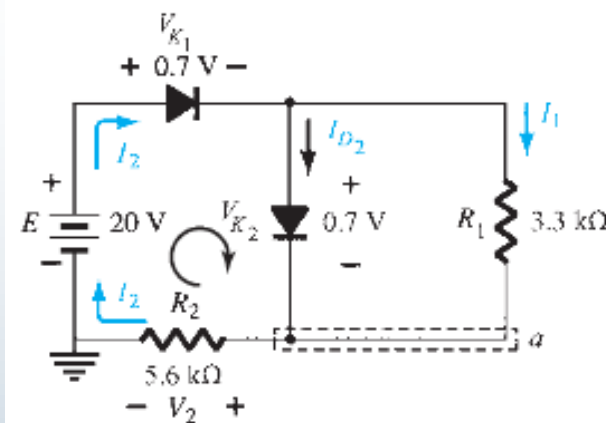
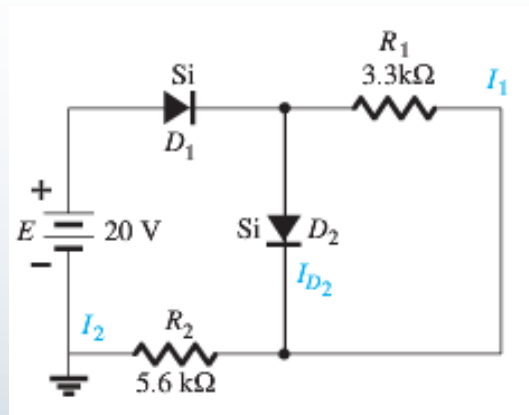
Find the resistor R to ensure a current of 20 mA through the “on” diode for the configuration of Fig. 2.30. Both diodes have a reverse breakdown voltage of 3 V and an average turn-on voltage of 2 V.



$$I = 20 \text{ mA} = \frac{E - V_{\text{LED}}}{R} = \frac{8 \text{ V} - 2 \text{ V}}{R}$$

$$R = \frac{6 \text{ V}}{20 \text{ mA}} = \mathbf{300 \, \Omega}$$

EXAMPLE 2.13 Determine the currents I_1 , I_2 , and I_{D_2} for the network of Fig. 2.37.



$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and
$$V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$$

with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$$

At the bottom node a ,

$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$$

2.5 AND/OR GATES

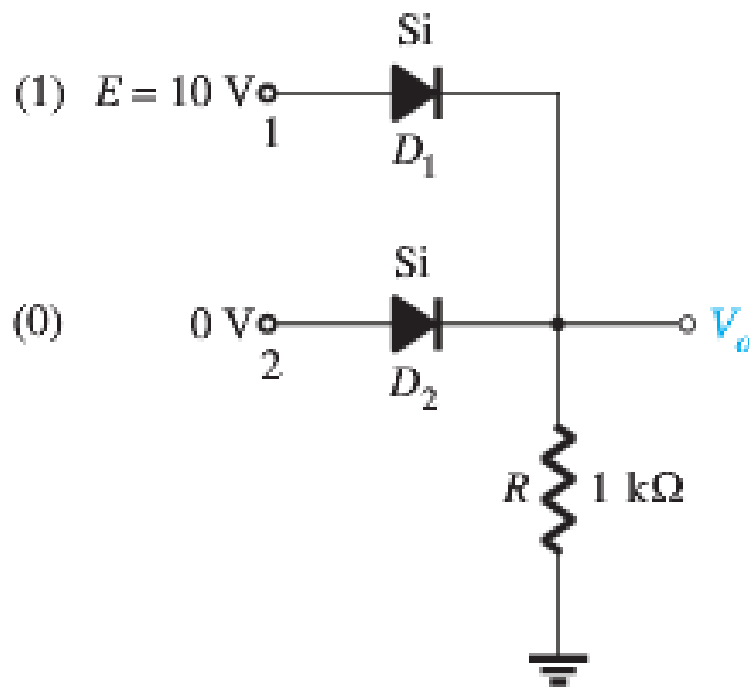


FIG. 2.39

Positive logic OR gate.

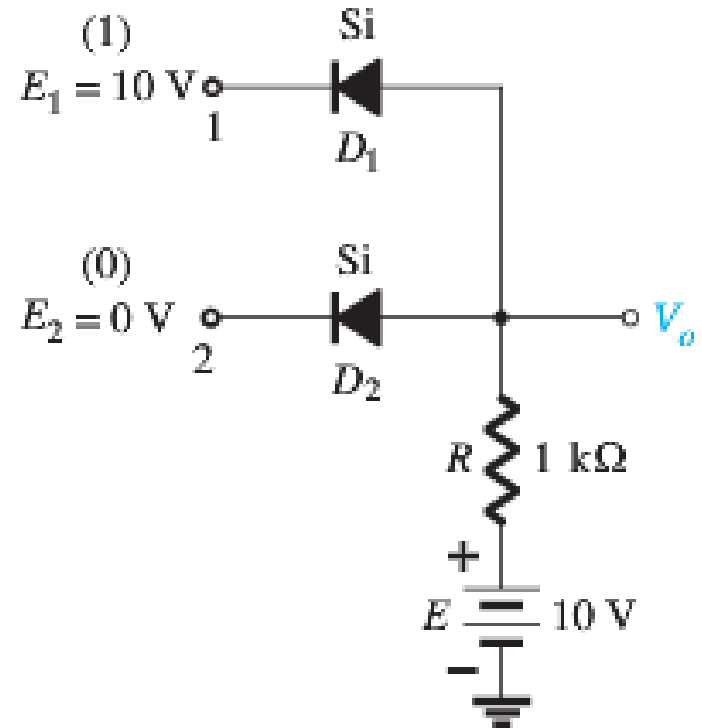


FIG. 2.42

Positive logic AND gate.

Quiz 2

- 24/06/2019: Monday
- Use 11th edition of the textbook
- Syllabus: 2.2, 2.3 and 2.4
- Examples:
- **Chapter 2:** 2.4 – 2.13
- Problems:
Chapter 2: 5, 6, 7, 8, 9, 10, 11, 12

Rectifier Circuits

- One important application of diode is the **rectifier**
 - Converts AC to DC
 - Used to make **dc power supplies**

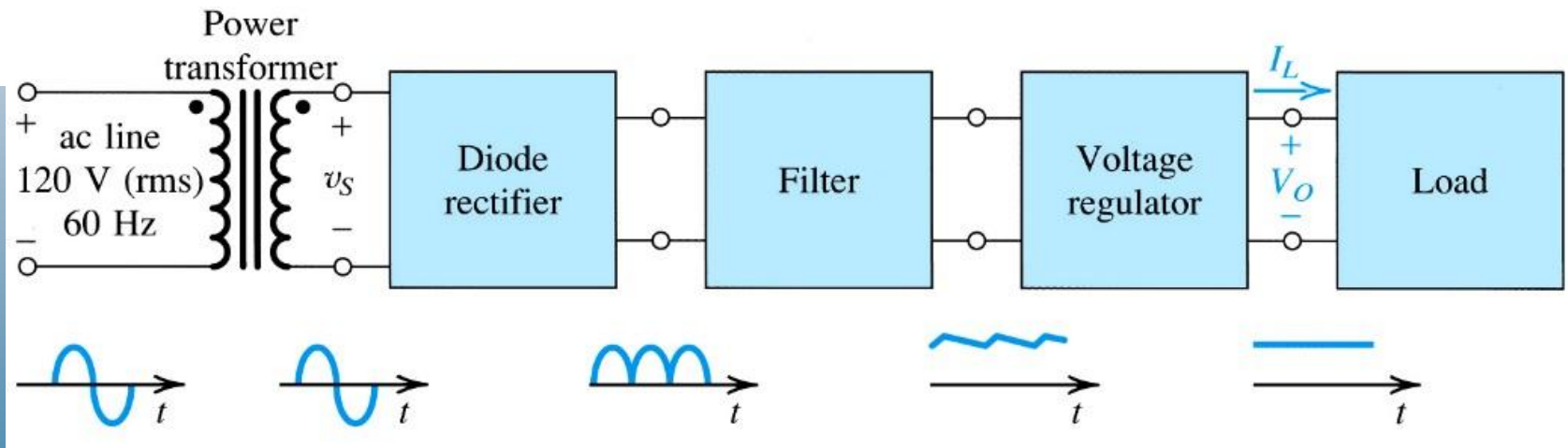
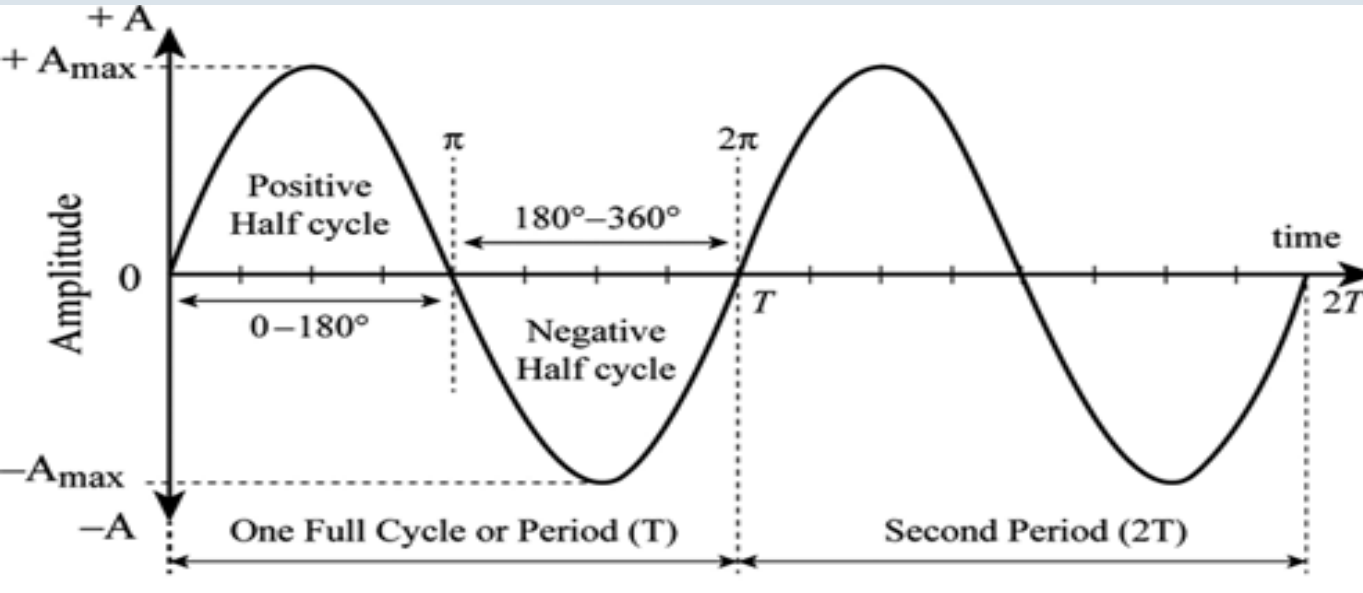


Figure: Block diagram of a dc power supply

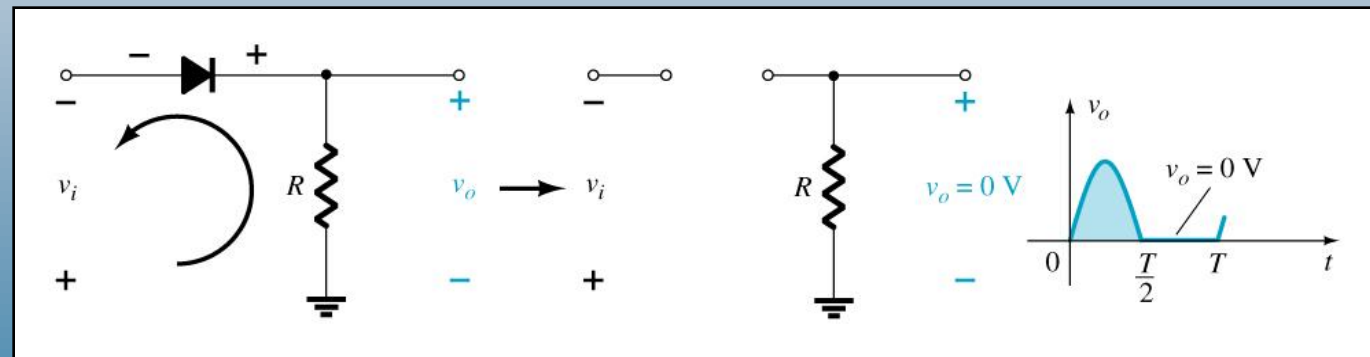
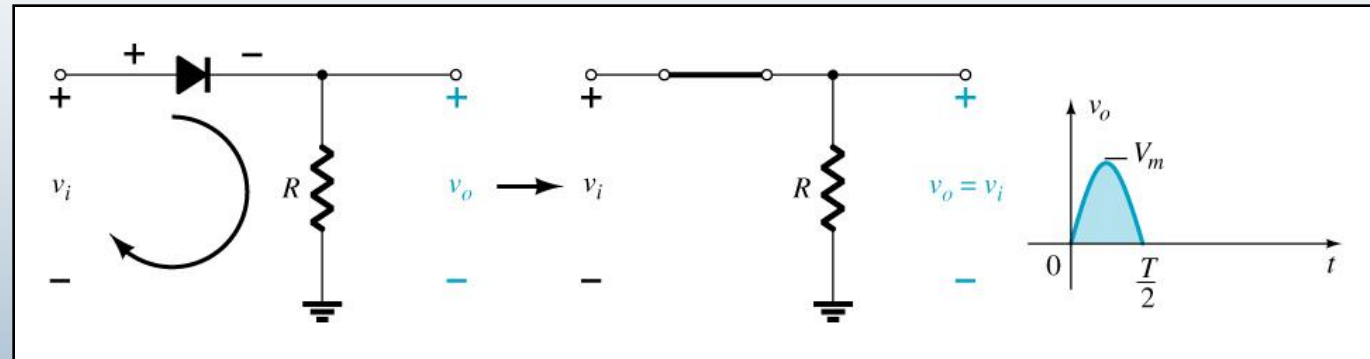
Input AC signal

- Rectifier input: Sine, square, triangular waves
- Sine wave: Positive and negative half cycle
- $v_{in} = V_m \sin(\omega t)$. $v_{in} = 10 \sin(\omega t) V$



2.6 Half-Wave Rectification

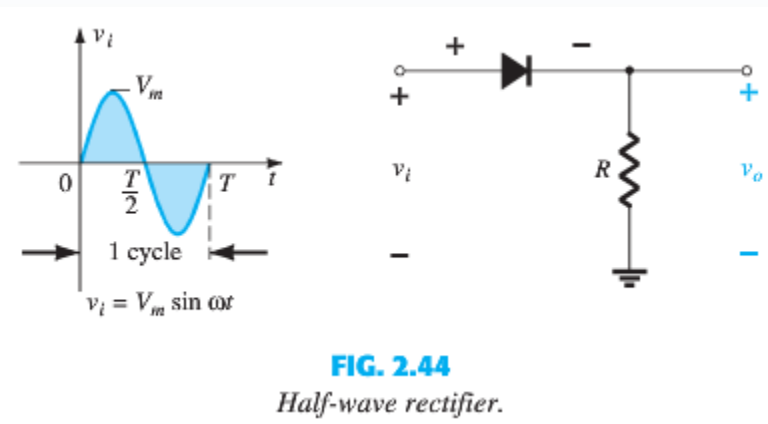
The diode conducts only when it is forward biased, therefore only half of the AC cycle passes through the diode to the output.



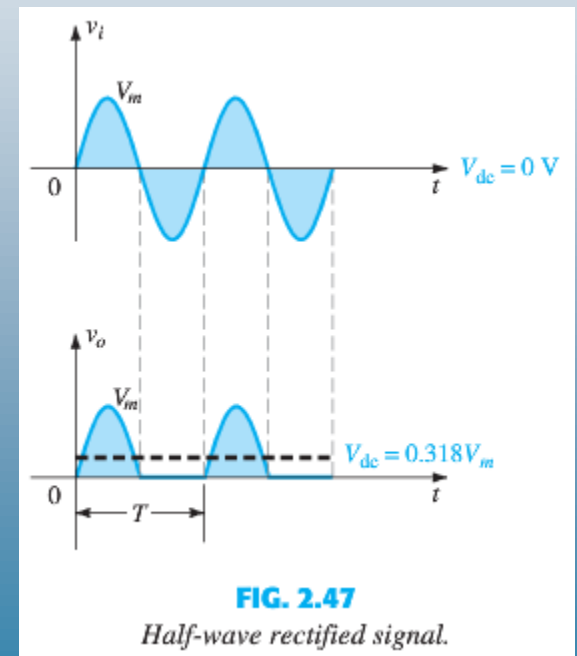
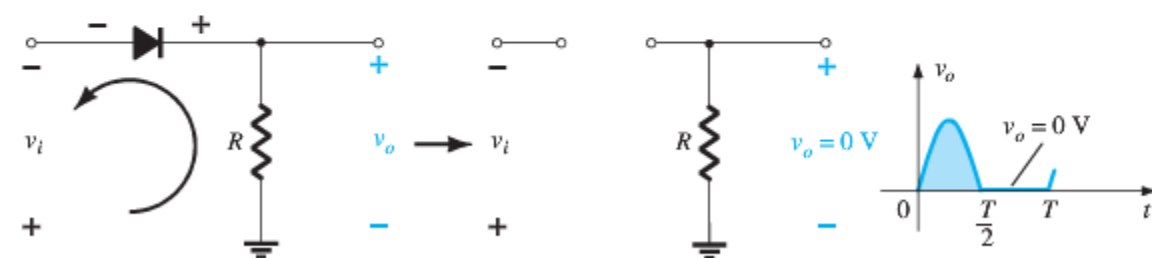
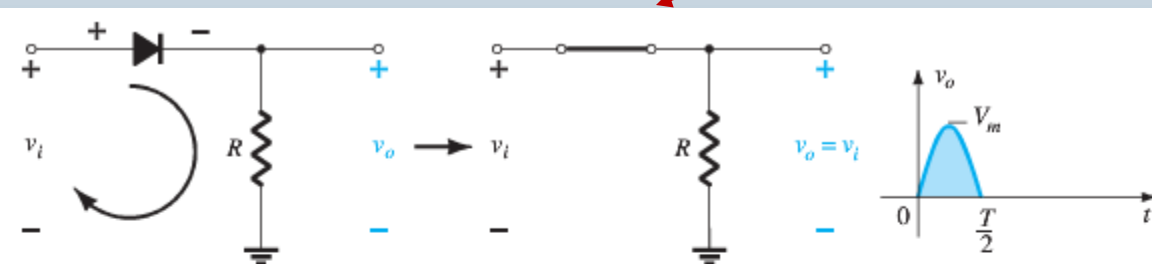
The DC output voltage is $0.318V_m$, where V_m = the peak AC voltage.

2.6 Half-Wave Rectification

The process of removing one-half the input signal to establish a dc level is called half-wave rectification



Ideal Approximation



$$V_{dc} = 0.318 V_m$$

half-wave

The effect of using a real diode

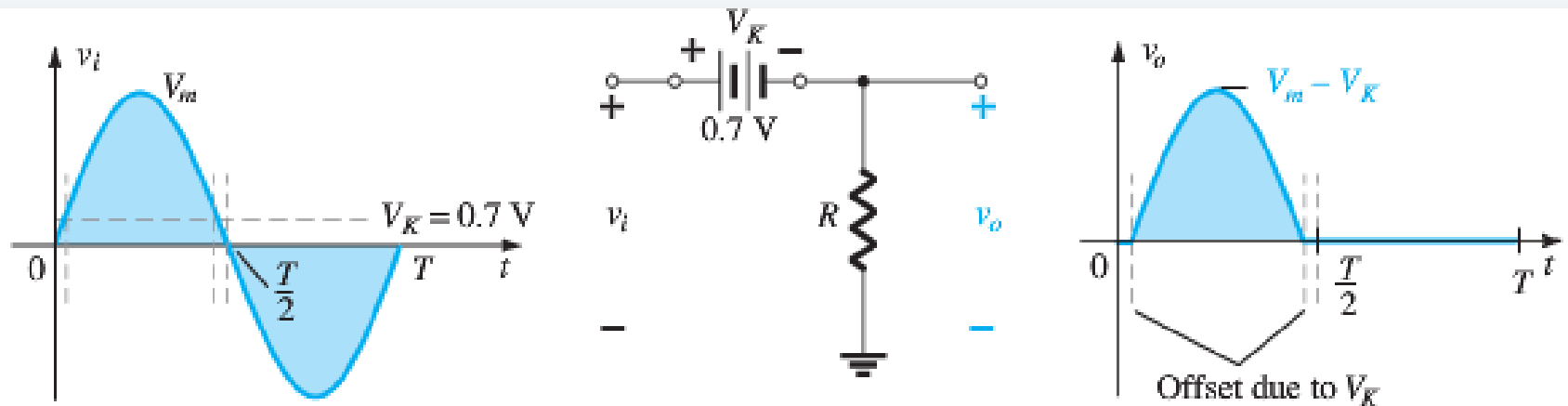


FIG. 2.48

Effect of V_K on half-wave rectified signal.

$$V_{\text{dc}} \cong 0.318(V_m - V_K)$$

The Half-Wave Rectifier

- Average Value (DC Component) of v_o , $V_o = \frac{\text{Integration over one cycle}}{\text{Length of a Period}} = \frac{\int_0^{2\pi} v_o d\theta}{2\pi}$

$$= \frac{\int_0^{\pi} (v_o - V_K) d\theta + \int_{\pi}^{2\pi} 0 d\theta}{2\pi}$$

$$= \frac{\int_0^{\pi} (V_m \sin \theta - V_K) d\theta}{2\pi}$$

$$= \frac{V_m}{2\pi} \int_0^{\pi} \sin \theta d\theta - \frac{V_K}{2\pi} \int_0^{\pi} d\theta$$

$$= \frac{V_m}{2\pi} [-\cos \theta]_0^{\pi} - \frac{V_K}{2\pi} [\theta]_0^{\pi}$$

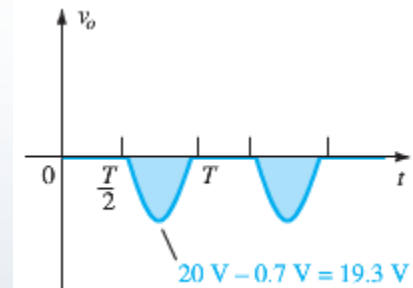
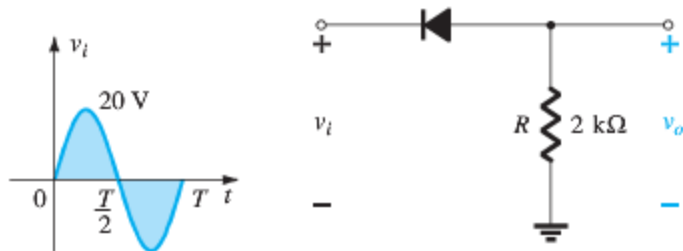
$$= -\frac{V_m}{2\pi} [\cos \theta]_0^{\pi} - \frac{V_K}{2\pi} \pi$$

$$= -\frac{V_m}{2\pi} (-2) - \frac{V_K}{2} = \frac{V_m}{\pi} - \frac{V_K}{2}$$

$$V_o = \frac{V_m}{\pi} - \frac{V_K}{2} = 0.318 V_m - 0.5 V_K$$
- Peak diode current $= (V_s - V_D) / R$

EXAMPLE 2.16

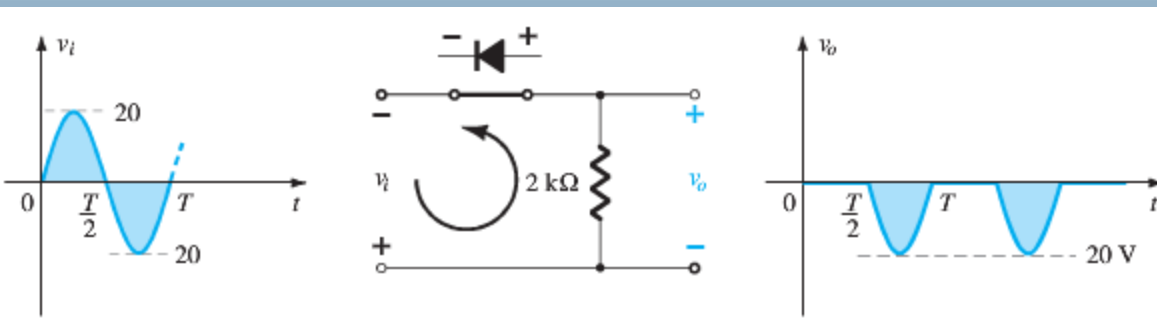
- Sketch the output v_o and determine the dc level of the output for the network of Fig. 2.49.
- Repeat part (a) if the ideal diode is replaced by a silicon diode.
- Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions using Eqs. (2.7) and (2.8).



$$V_{\text{dc}} \cong -0.318(V_m - 0.7\text{ V}) = -0.318(19.3\text{ V}) \cong -6.14\text{ V}$$

- In this situation the diode will conduct during the negative part of the input as shown in Fig. 2.50, and v_o will appear as shown in the same figure. For the full period, the dc level is

$$V_{\text{dc}} = -0.318V_m = -0.318(20\text{ V}) = -6.36\text{ V}$$



$$V_{\text{dc}} = -0.318 V_m = -0.318(200\text{ V}) = -63.6\text{ V}$$

$$V_{\text{dc}} = -0.318(V_m - V_K) = -0.318(200\text{ V} - 0.7\text{ V}) \\ = -(0.318)(199.3\text{ V}) = -63.38\text{ V}$$

which is a difference that can certainly be ignored for most applications.

PIV (PRV)

The diode is only forward biased for one-half of the AC cycle. It is also reverse biased for one-half cycle.

It is important that the reverse breakdown voltage rating of the diode be high enough to withstand the peak, reverse-biasing AC voltage.

$$\text{PIV (or PRV)} > V_m$$

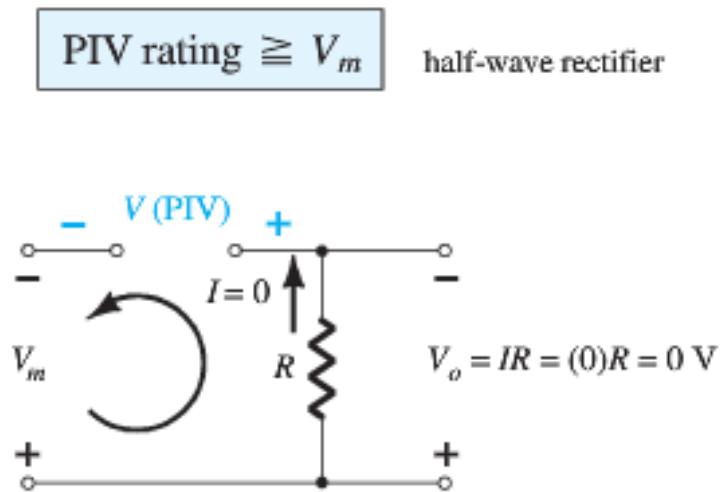


FIG. 2.52

Determining the required PIV rating for the half-wave rectifier.

Where **PIV** = Peak inverse voltage

PRV = Peak reverse voltage

V_m = Peak AC voltage

2.7 Full-Wave Rectification

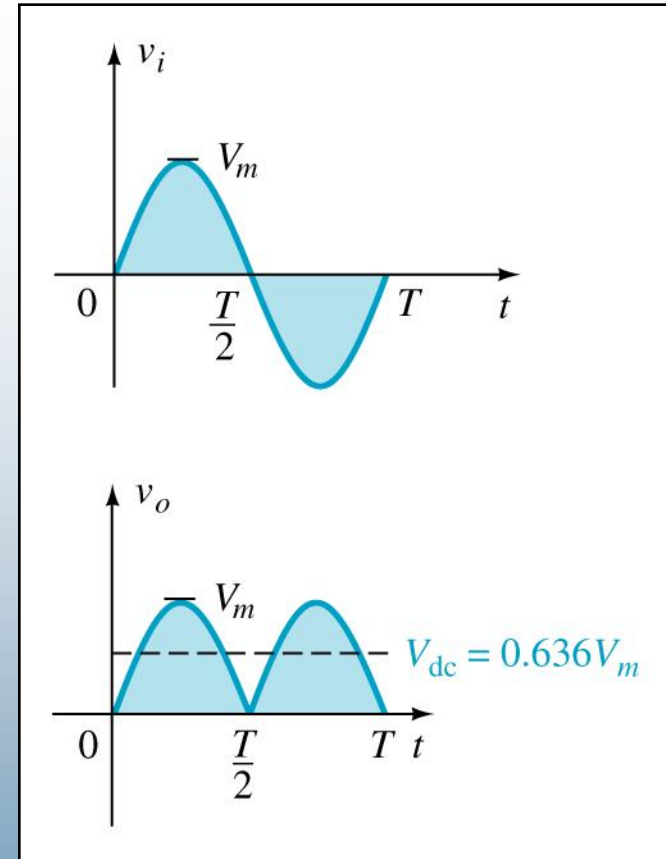
The rectification process can be improved by using a full-wave rectifier circuit.

Full-wave rectification: output for both positive and negative half cycle

Full-wave rectification produces a larger DC output:

$$\text{Half-wave: } V_{dc} = \frac{V_m}{\pi} = 0.318V_m$$

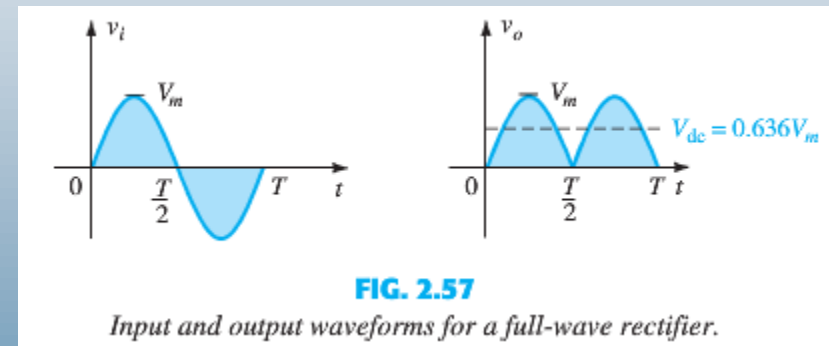
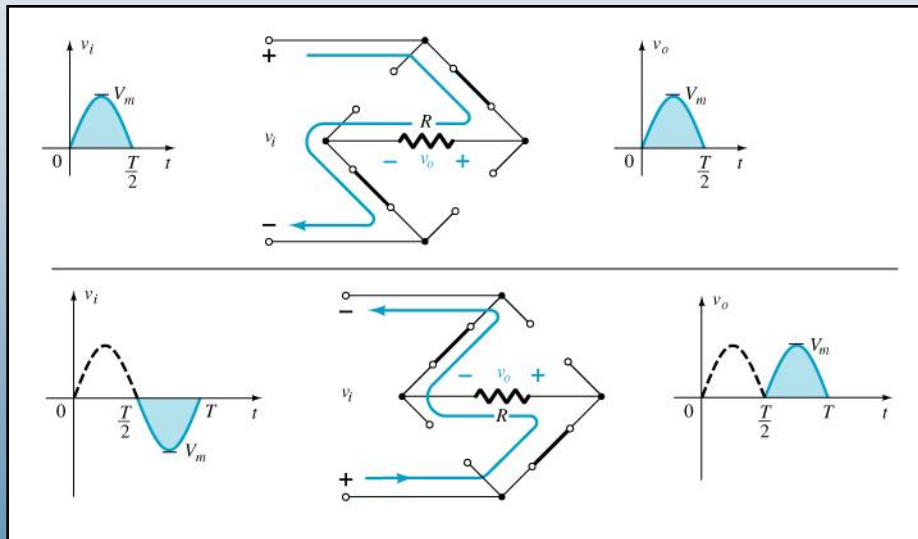
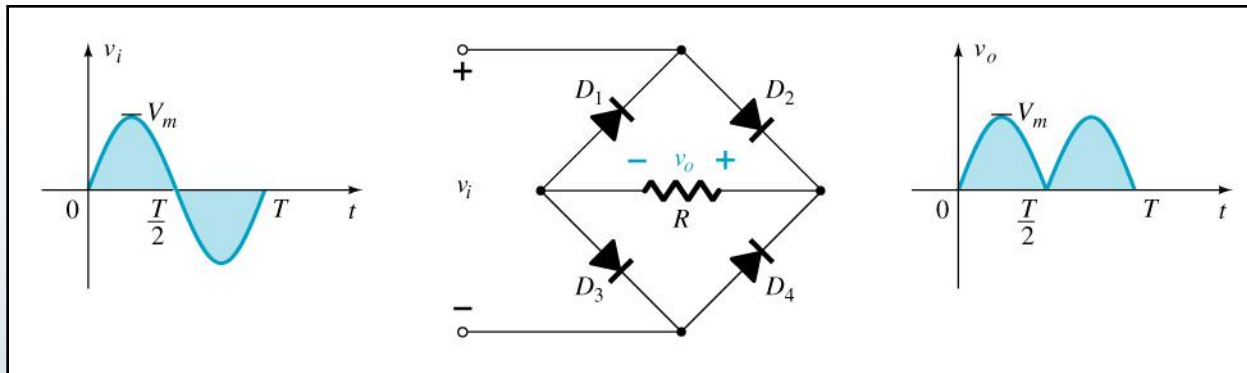
$$\text{Full-wave: } V_{dc} = \frac{2*V_m}{\pi} = 0.636V_m$$



Full-Wave Rectification

Bridge Rectifier

A full-wave rectifier with four diodes that are connected in a bridge configuration



$$V_{dc} = 2[\text{Eq. (2.7)}] = 2(0.318V_m)$$

$$V_{dc} = 0.636 V_m \quad \text{full-wave}$$

The effect of using a real diode

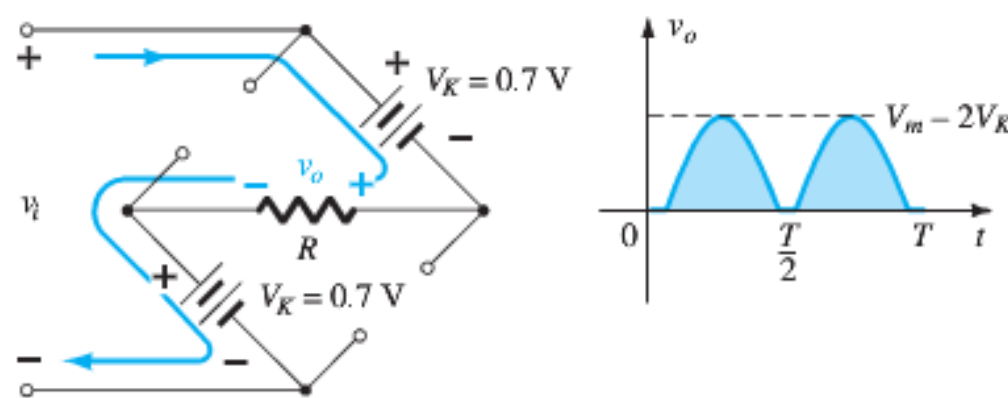


FIG. 2.58

Determining $V_{o_{\max}}$ for silicon diodes in the bridge configuration.

$$v_i - V_K - v_o - V_K = 0$$

$$v_o = v_i - 2V_K$$

The peak value of the output voltage v_o is therefore

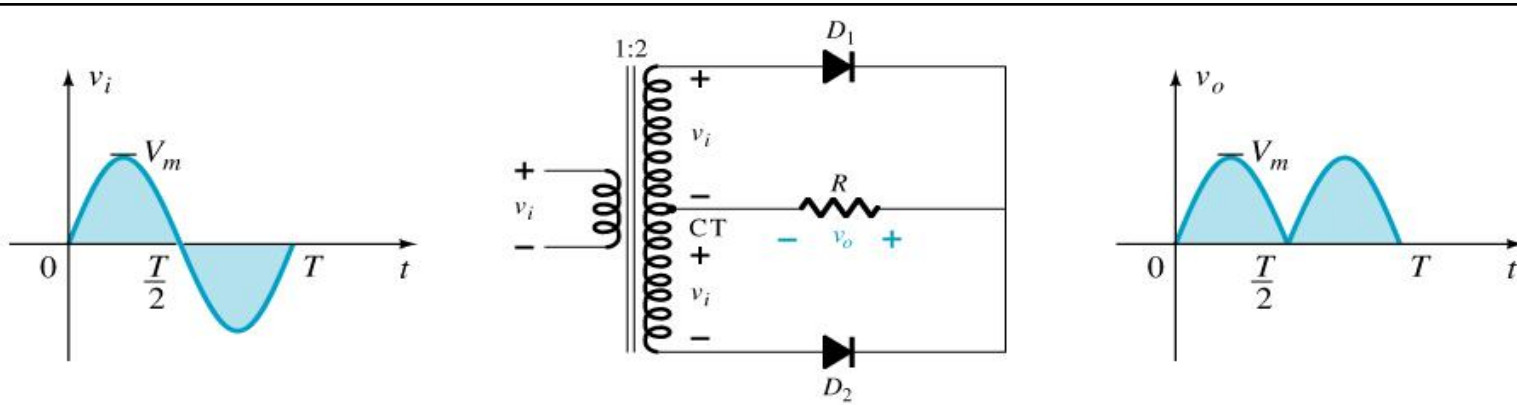
$$V_{o_{\max}} = V_m - 2V_K$$

For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$V_{\text{dc}} \cong 0.636(V_m - 2V_K) \quad (2.11)$$

Then again, if V_m is sufficiently greater than $2V_K$, then Eq. (2.10) is often applied as a first approximation for V_{dc} .

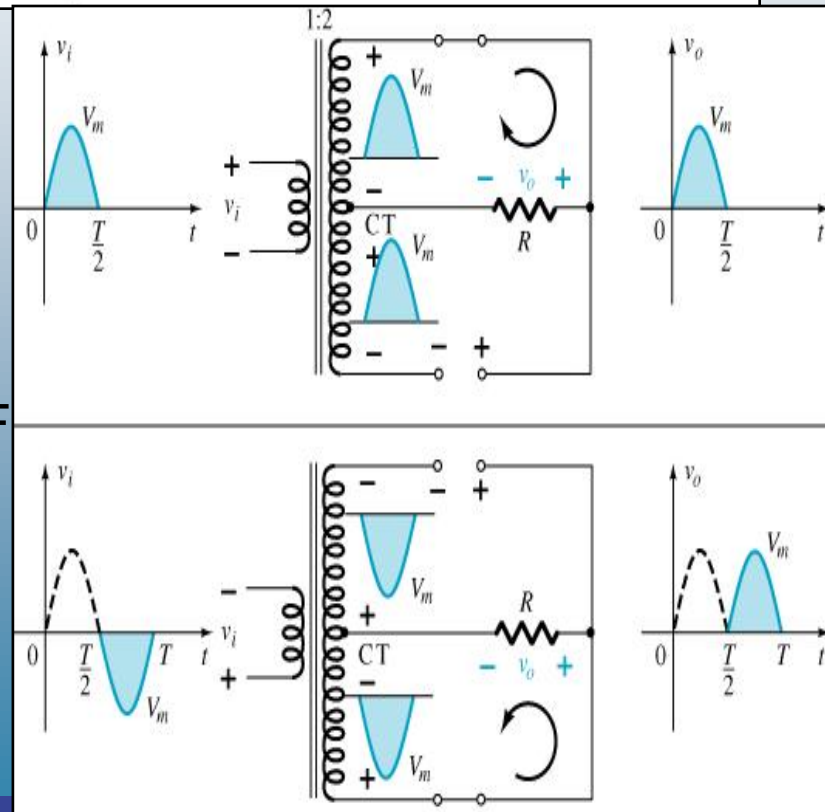
Full-Wave Rectification



Center-Tapped Transformer Rectifier

- Requires two diodes and a center-tapped transformer.
- (+) half cycle: D_1 ON; D_2 OFF
- (-) half cycle: D_1 OFF; D_2 ON

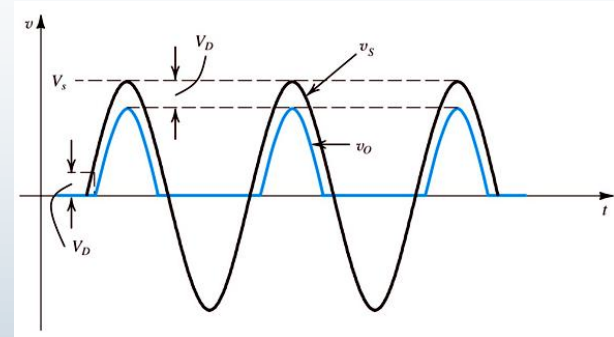
$$V_{DC} = 0.636V_m$$



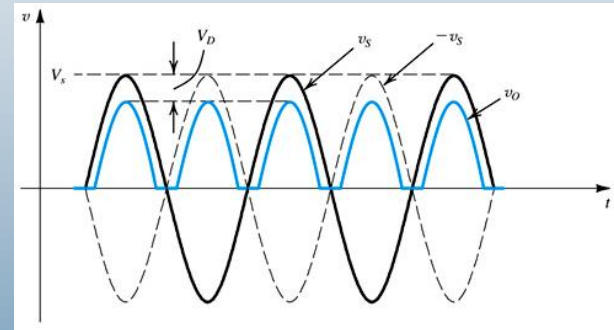
Needs a bigger transformer, because the peak is only half the secondary voltage.

Summary of Rectifier Waveforms

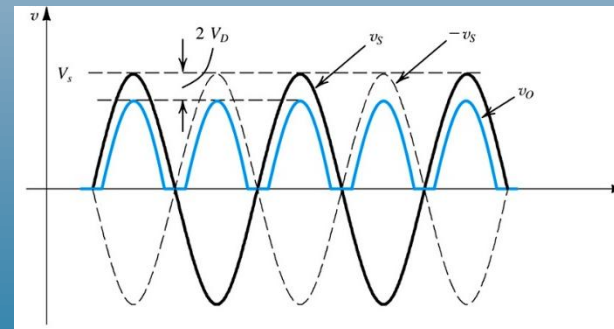
Half-Wave Rectifier



Full-Wave Rectifier



Bridge Rectifier



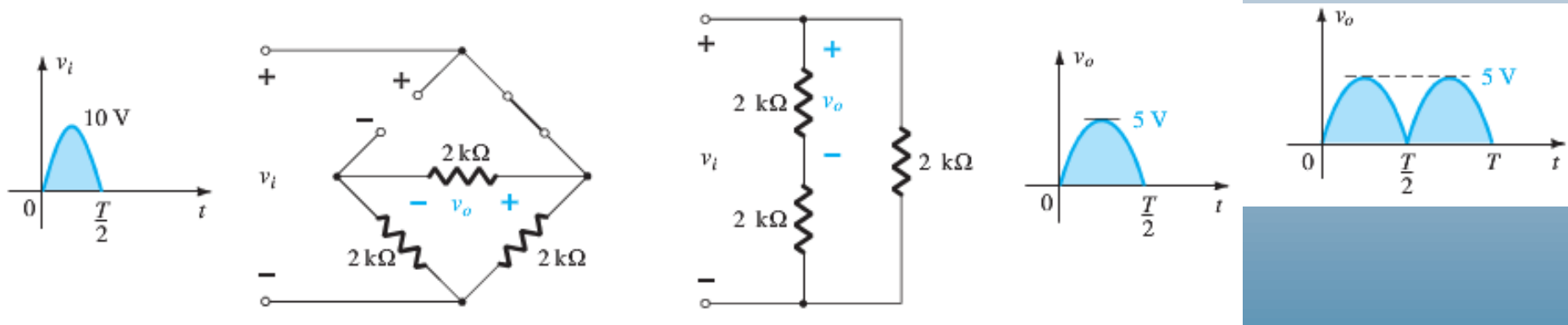
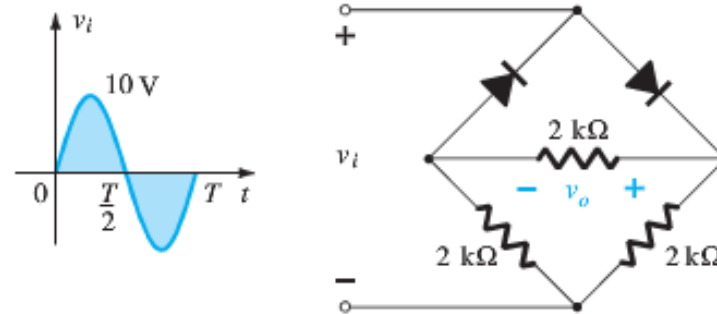
Summary of Rectifier Circuits

In the center tapped transformer rectifier circuit, the peak AC voltage is the transformer secondary voltage to the tap.

Rectifier	Ideal V_{DC}	Realistic Si V_{DC}
Half Wave Rectifier	$V_{DC} = 0.318V_m$	$V_{DC} = 0.318(V_m - 0.7V)$
Bridge Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636(V_m - 2(0.7 V))$
Center-Tapped Transformer Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636(V_m - 0.7 V)$

V_m = the peak AC voltage

EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.



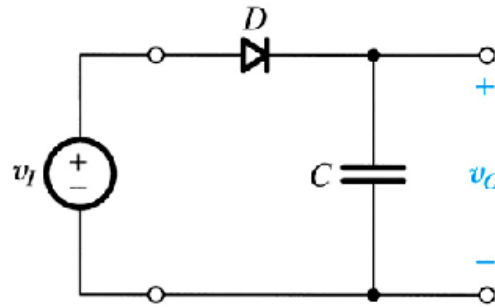
$$V_{o_{\max}} = \frac{1}{2}V_{i_{\max}} = \frac{1}{2}(10\text{ V}) = 5\text{ V}$$

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

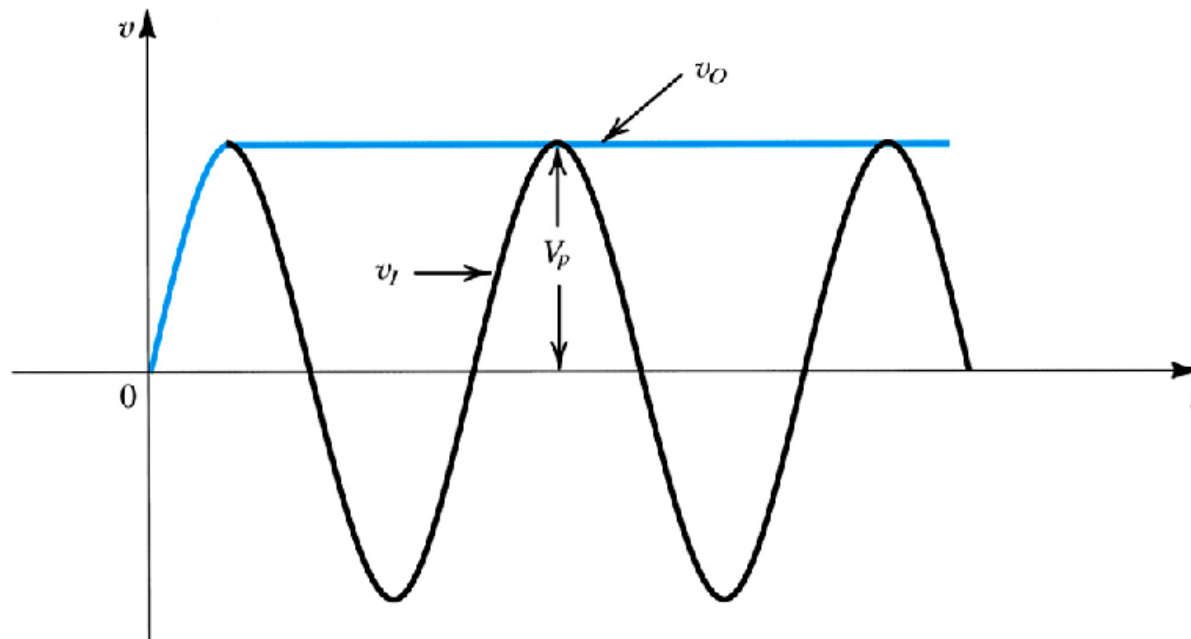
$$V_{\text{dc}} = 0.636(5\text{ V}) = \mathbf{3.18\text{ V}}$$

The Rectifier with a Filter Capacitor

- The Peak Rectifier

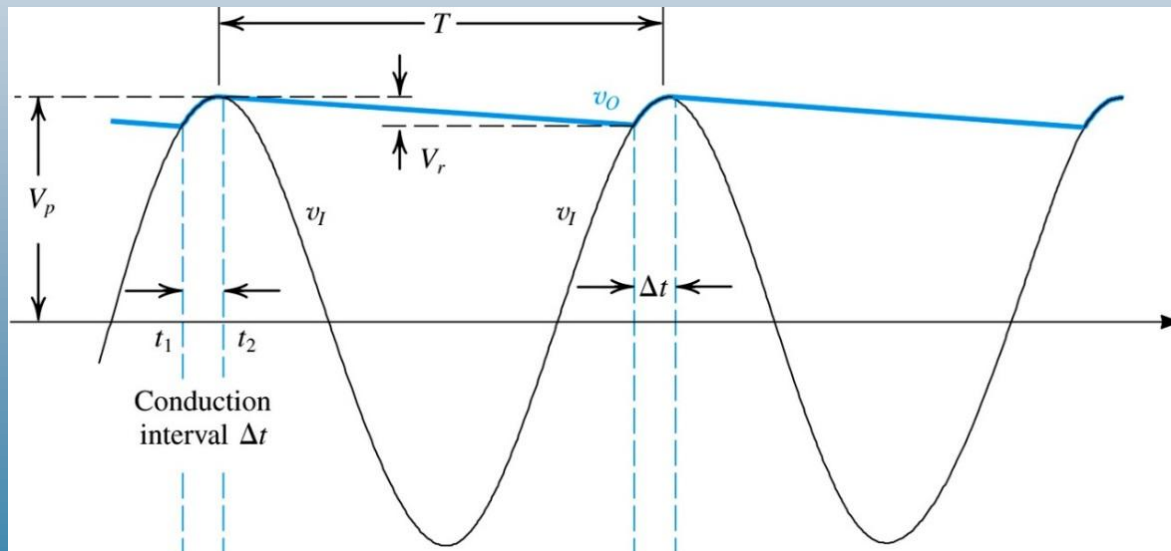
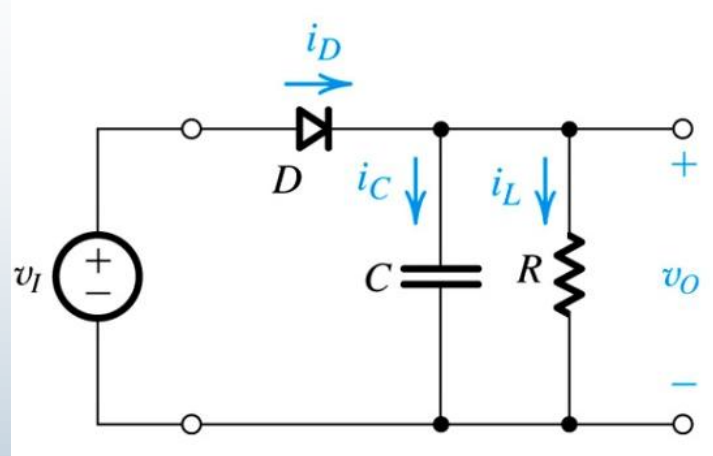


(a)

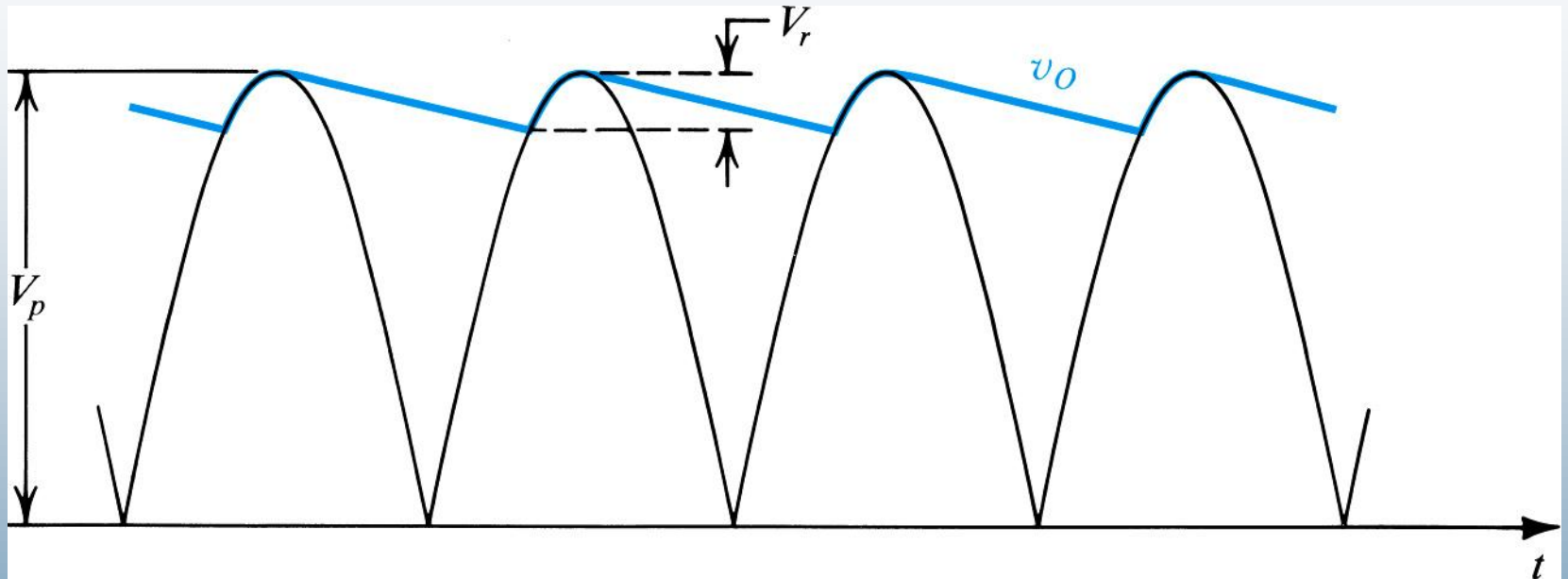


(b)

The Peak Rectifier with a Load Resistor – A More Practical Circuit



The full-wave peak rectifier



- The ripple is smaller than half-wave rectifier
- $V_r = \frac{V_p}{C * R_L}$
- Ripple voltage inversely proportional to filter capacitor

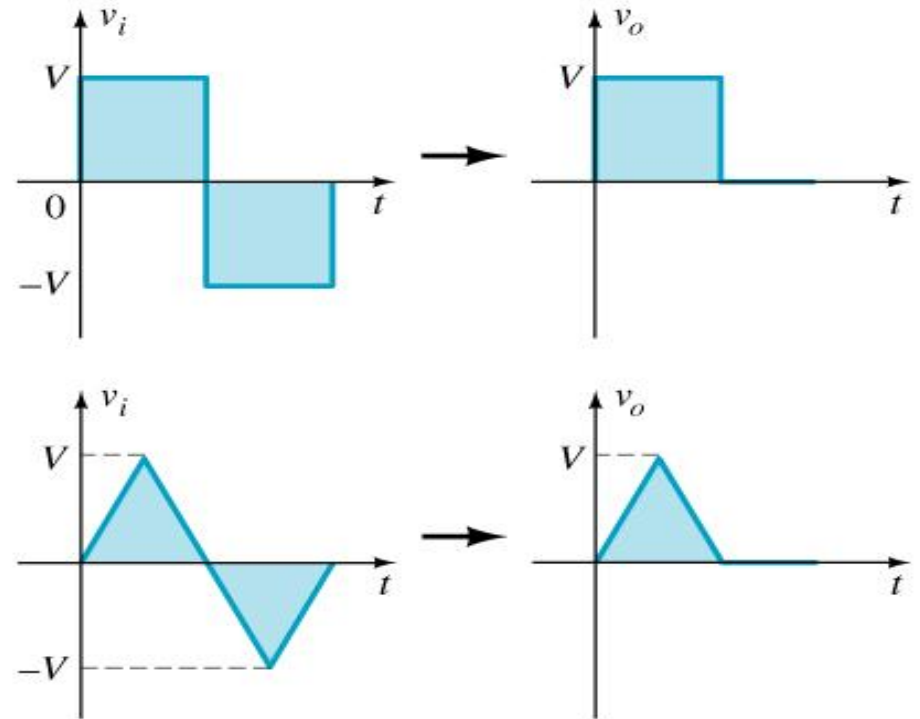
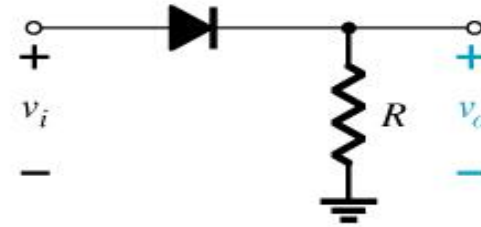
2.8 Diode Clippers

- Clippers are networks that employ diodes to “clip” away or removes a portion of an input signal without distorting the remaining part of the applied waveform.
- The half-wave rectifier is an example of the simplest form of diode clipper—one resistor and a diode.
- Depending on the orientation of the diode, the positive or negative region of the applied signal is “clipped” off.
- Two general categories of clippers: series and parallel.
 - Series clippers: The diode in series with the load
 - Parallel clippers: The diode in parallel to the load

Diode Series Clippers

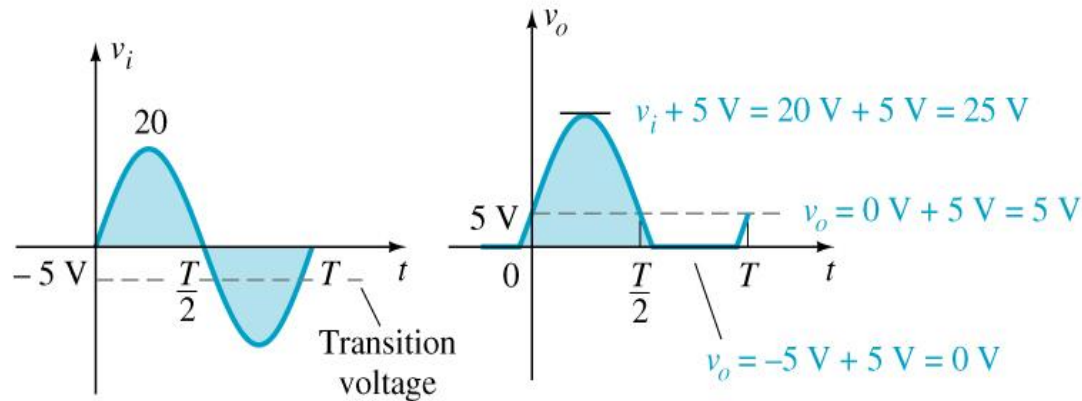
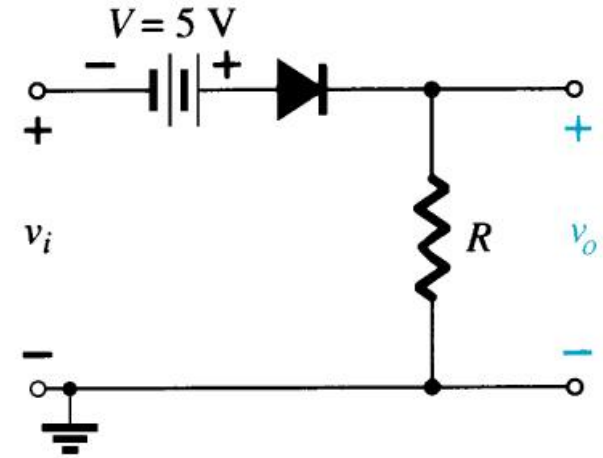
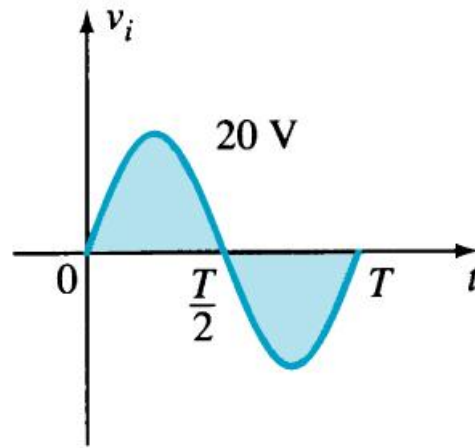
The diode in a series clipper “clips” any voltage that does not forward bias it:

- Series Clipper: Diode and Load resistance (R_L) in series
- Clips negative portion of the input: **Negative Clipper**
- A reverse-biasing polarity
- A forward-biasing polarity less than 0.7 V (for a silicon diode)



Biased Clippers

Adding a DC source in series with the clipping diode changes the effective forward bias of the diode.

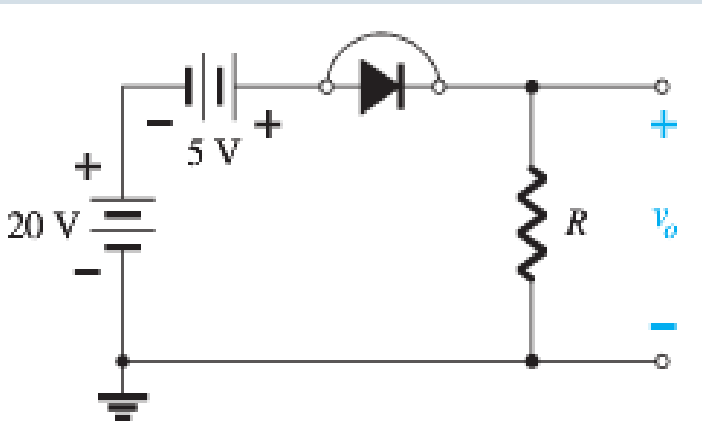
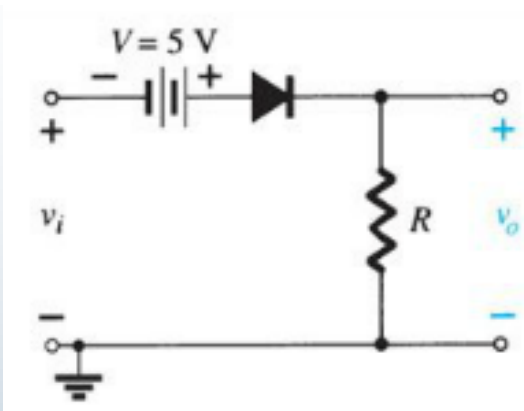
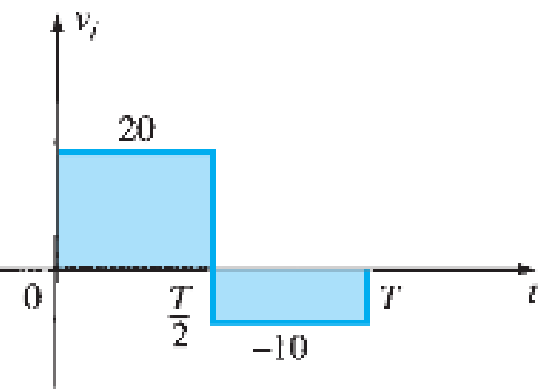


Biased Clipper: A DC source also present in the circuit

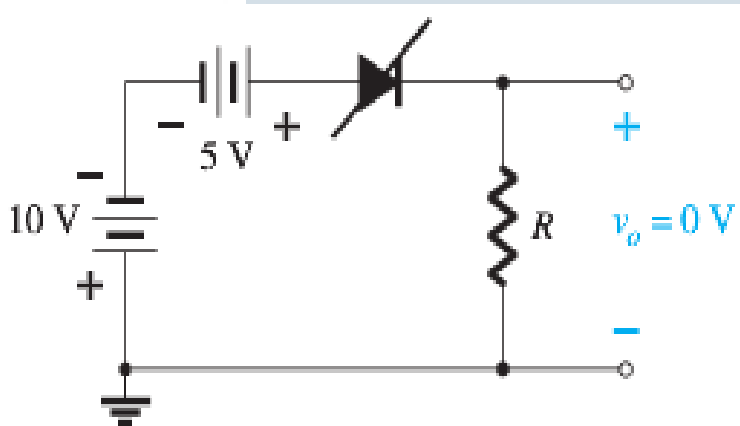
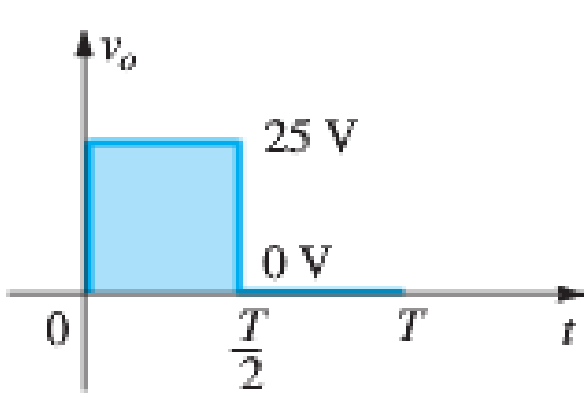
Series Clipper

Negative Clipper

EXAMPLE 2.19 Find the output voltage for the network examined in Example 2.18 if the applied signal is the square wave of Fig. 2.77.



Positive half cycle
 Diode FB/ON/Short
 KVL: $-20V - 5V + V_o = 0$
 $V_o = 25V$



Negative half cycle
 Diode RB/OFF/Open
 $I_D = 0A$
 $V_o = 0V$

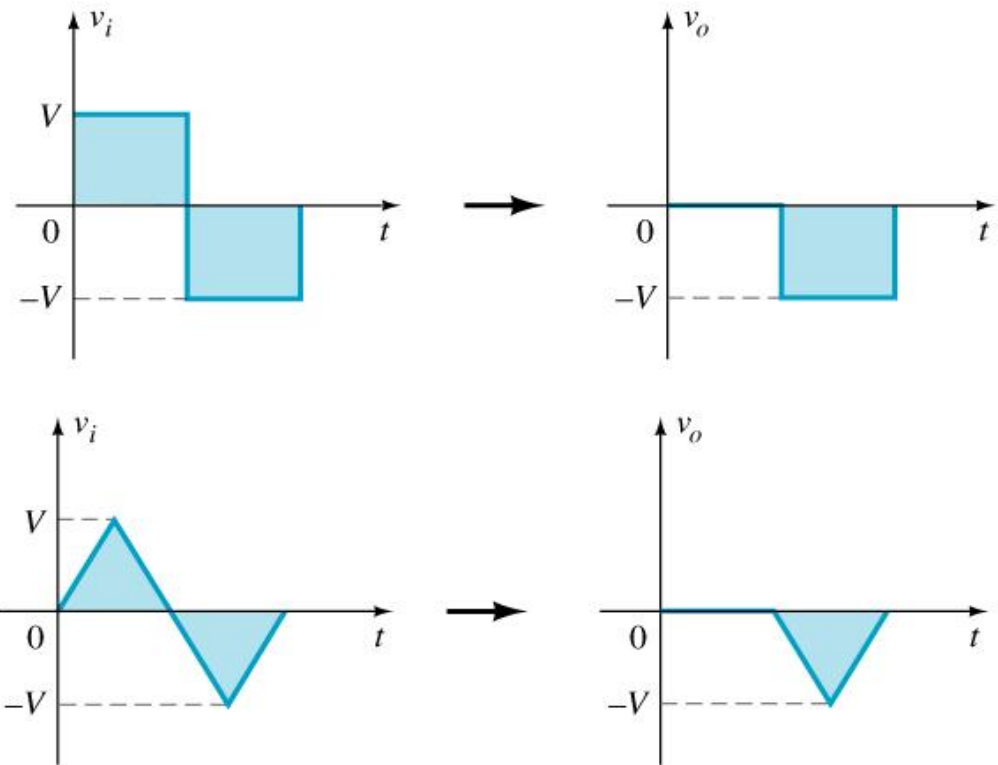
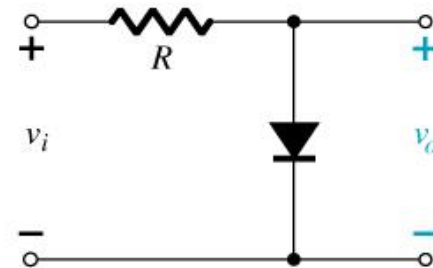
Parallel Clippers

The diode in a parallel clipper circuit “clips” any voltage that forward biases it.

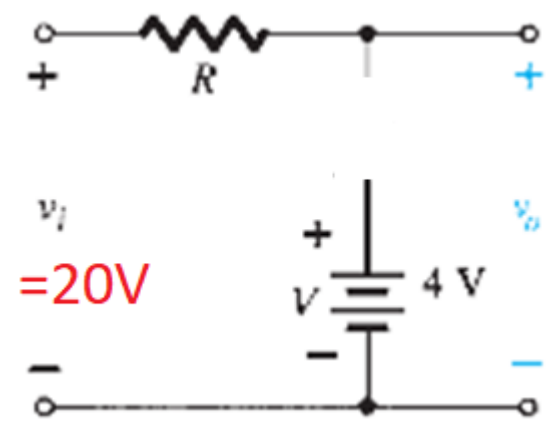
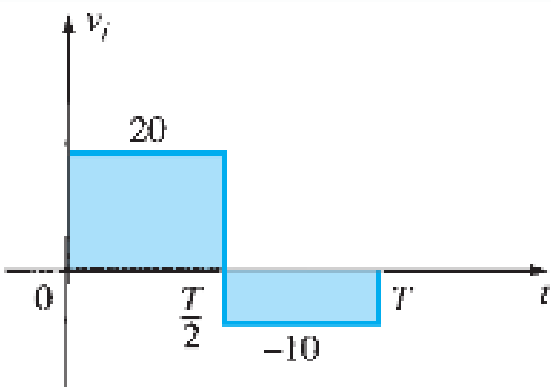
DC biasing can be added in series with the diode to change the clipping level.

Parallel Clipper: Diode and Load resistance (R_L) in parallel

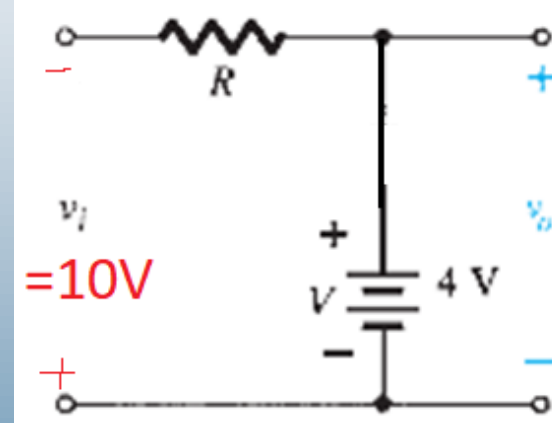
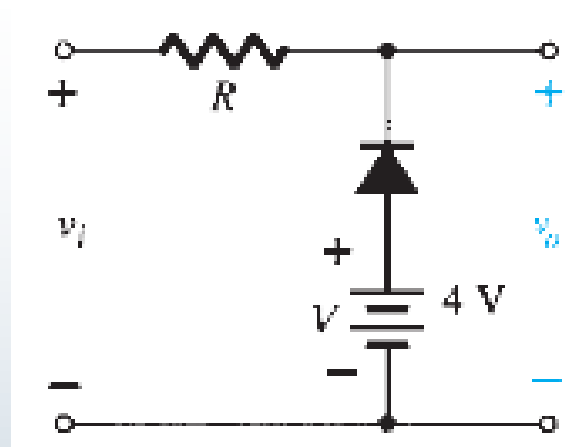
Clips positive portion of the input:
Positive Clipper



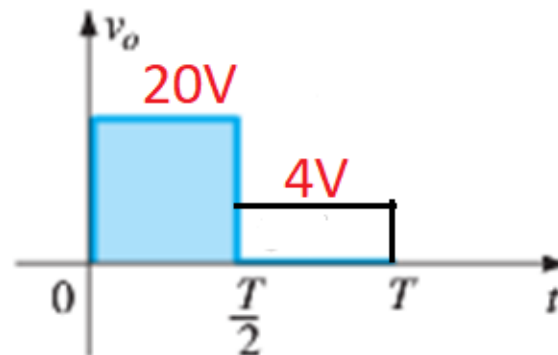
EXAMPLE 2.20 Determine v_o for the network of Fig. 2.82.



Positive half cycle
Diode RB/OFF/Open
 $V_o = V_{in} = 20V$



Negative half cycle
Diode FB/ON/Short
 $V_o = 4V$



EXAMPLE 2.21 Repeat Example 2.20 using a silicon diode with $V_K = 0.7$ V.

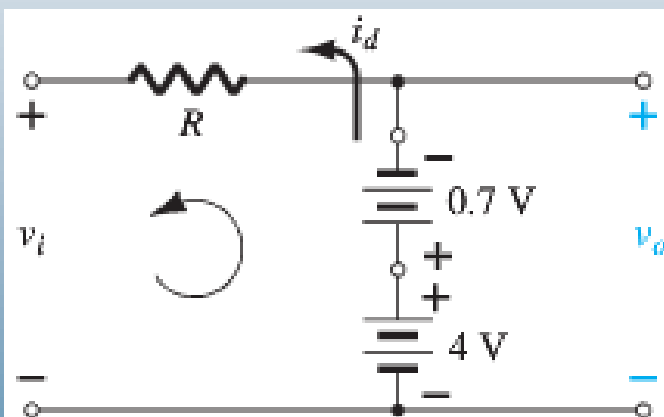
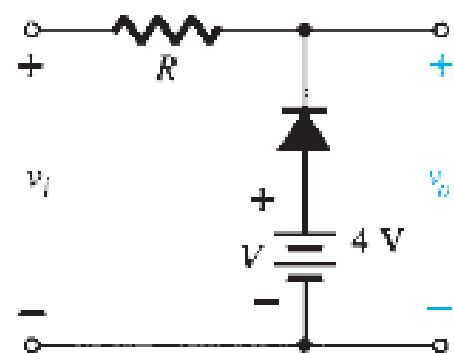
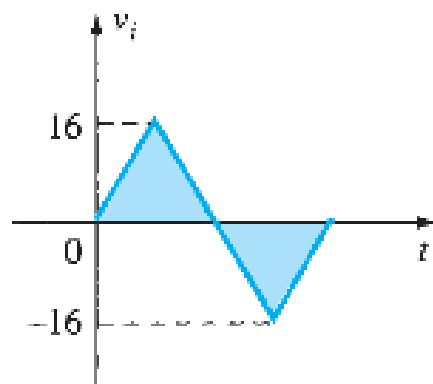


FIG. 2.86

Determining v_o for the diode of Fig. 2.82 in the "on" state.

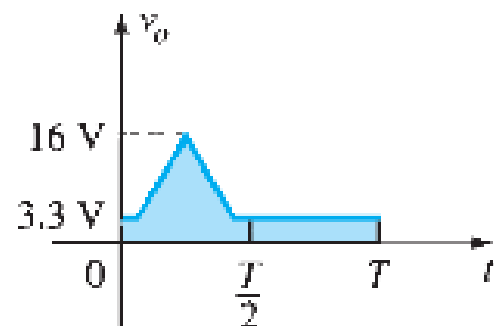


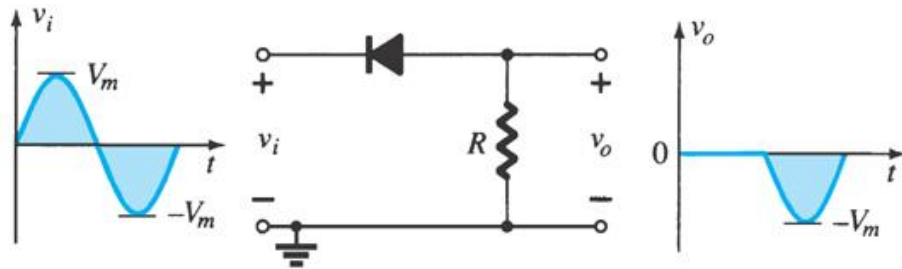
FIG. 2.87

Sketching v_o for Example 2.21.

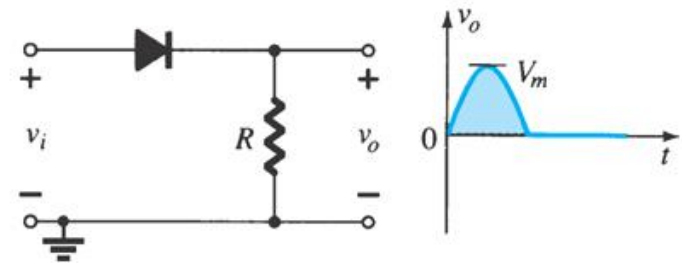
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

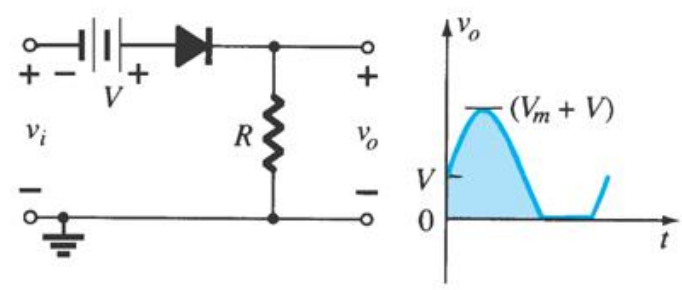
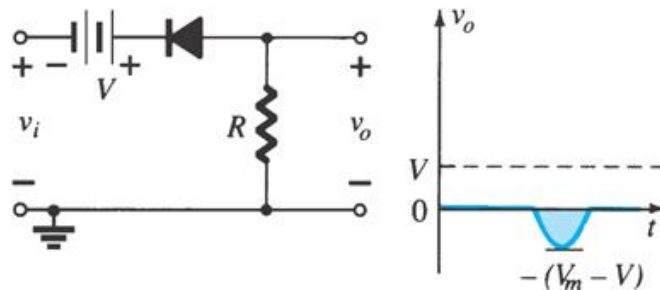
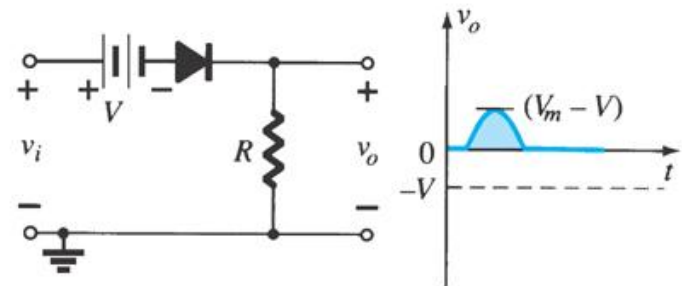
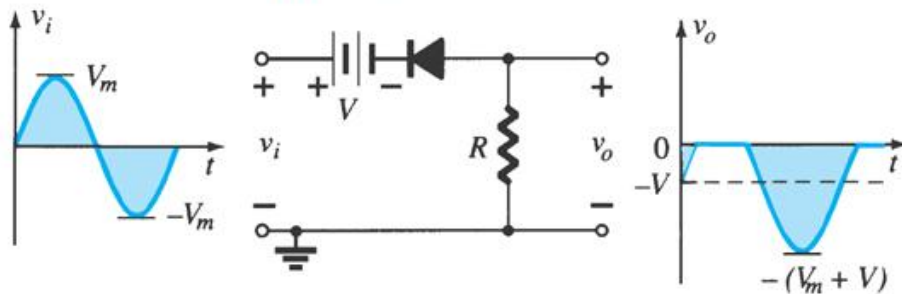
POSITIVE



NEGATIVE

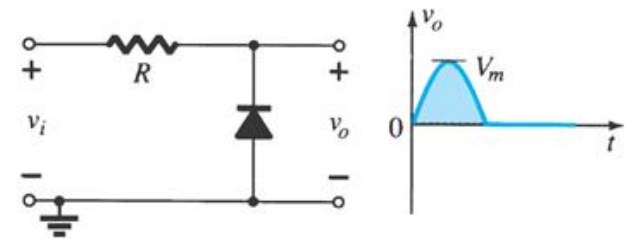
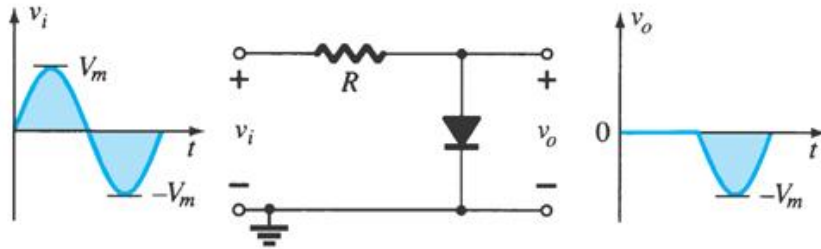


Biased Series Clippers (Ideal Diodes)

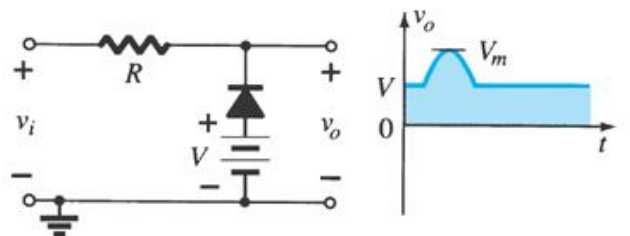
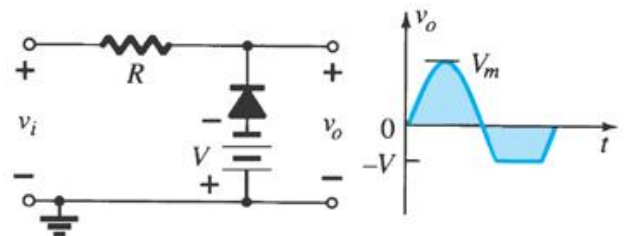
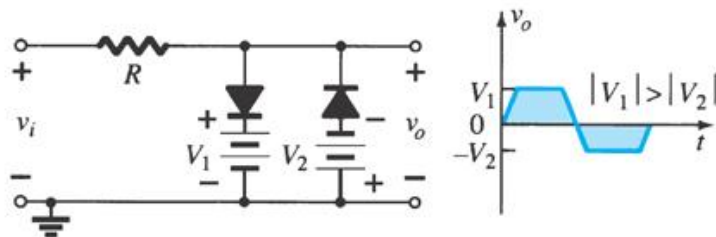
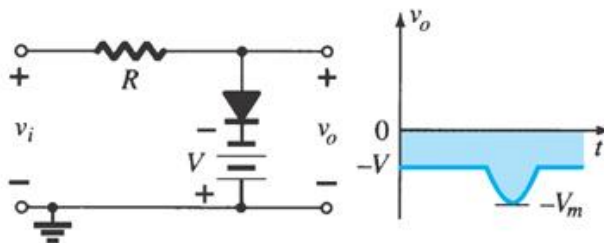
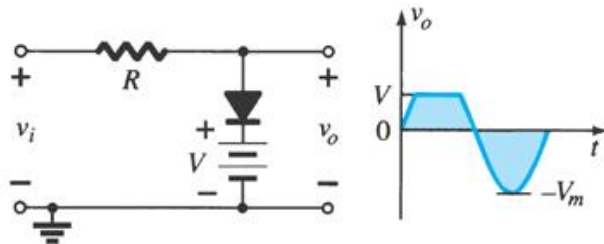


Summary of Clipper Circuits

Simple Parallel Clippers (Ideal Diodes)

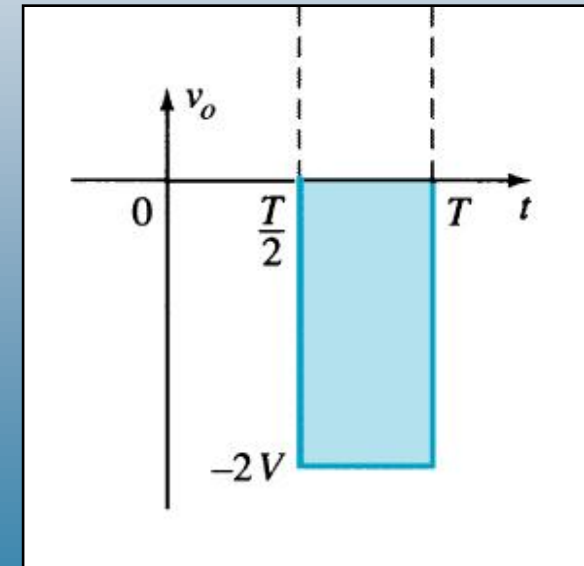
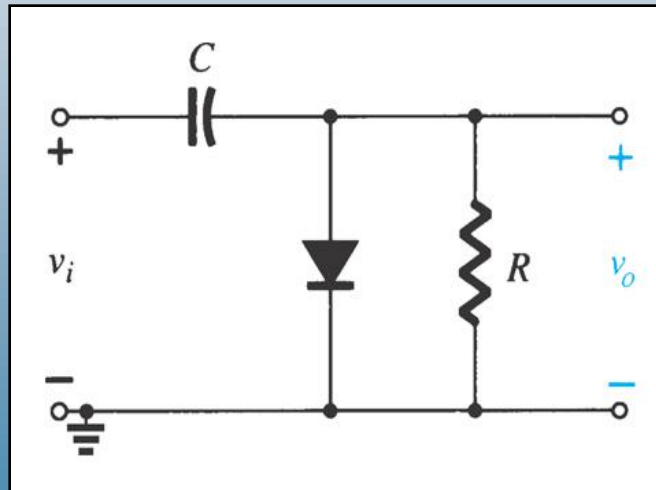
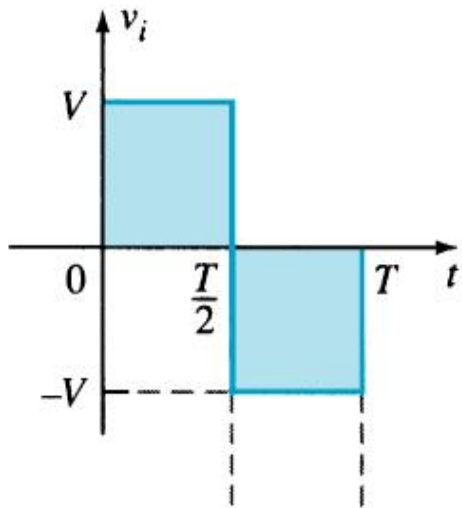


Biased Parallel Clippers (Ideal Diodes)

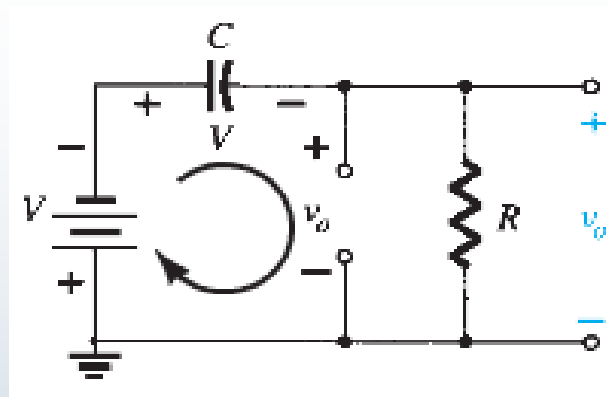
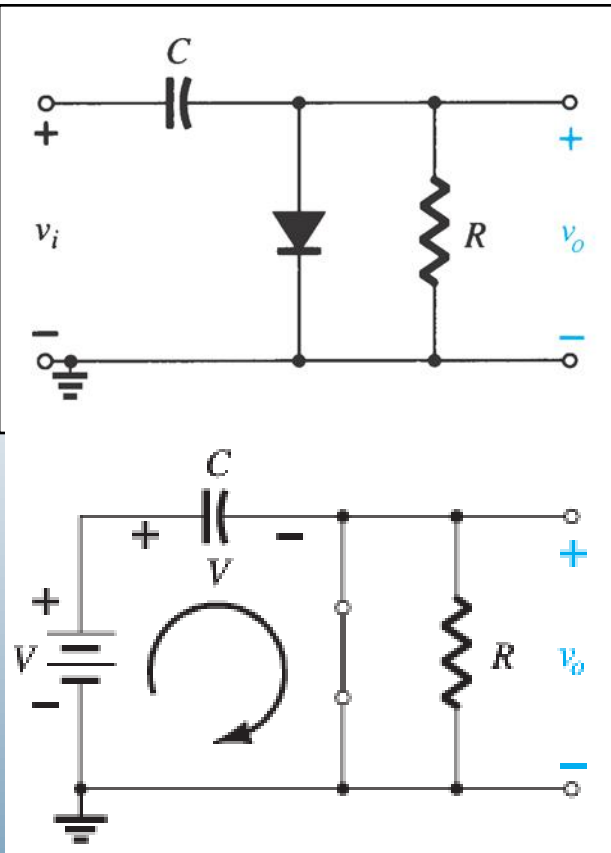


2.9 Clampers

- A clamper is a network constructed of a diode, a resistor, and a **capacitor** that **shifts a waveform to a different dc level** without changing the appearance of the applied signal.
- Clamping networks have a **capacitor** connected directly from input to output with a resistive element in parallel with the output signal.
 - The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.
 - **Peak-peak or swing of input and output must be same.**



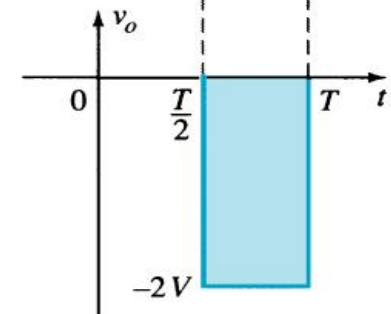
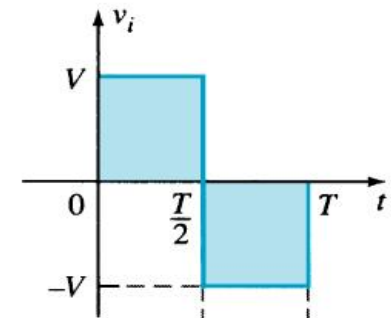
2.9 Clampers



1. **Negative half cycle:** Capacitor discharges. Capacitor voltage and polarity will not change. Diode RB/OFF. Find output (v_o).

$$v_o \neq 0V.$$

KVL: $+V_{in} + V + v_o = 0 \leftrightarrow v_o = -2V_{in}$.
Peak-peak or swing of input and output must be same



1. Check in which cycle diode is FB/ON?

Positive half cycle.

1. **Positive half cycle:** Capacitor charges.

Diode FB/ON.

Find capacitor voltage (V). Find output (v_o).

$$v_o = 0V.$$

KVL: $-V_{in} + V = 0 \leftrightarrow V = V_{in}$.

EXAMPLE 2.22 Determine v_o for the network of Fig. 2.93 for the input indicated.

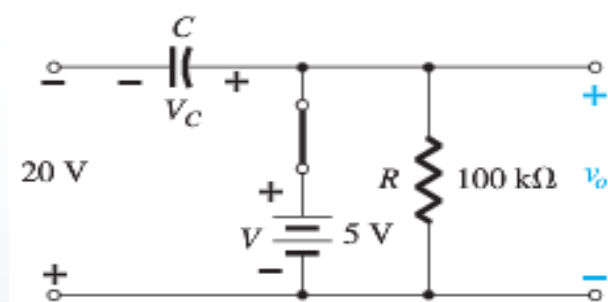
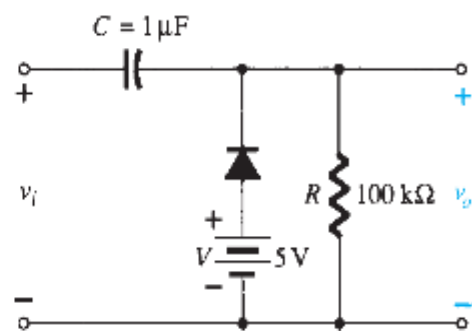
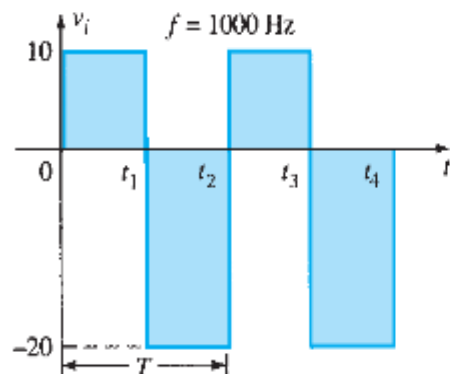


FIG. 2.94

Determining v_o and V_C with the diode in the “on” state.

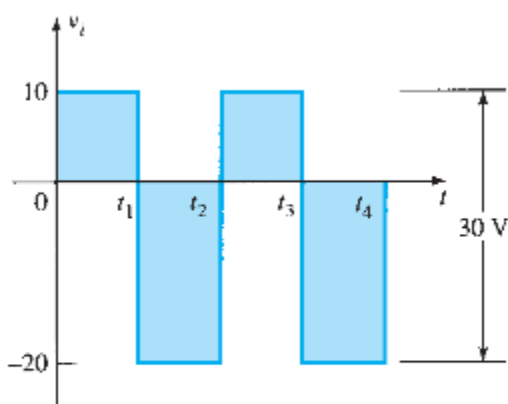


FIG. 2.96

v_i and v_o for the clamper of Fig. 2.93.

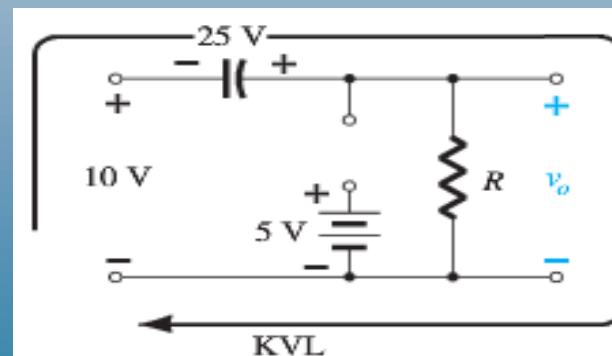
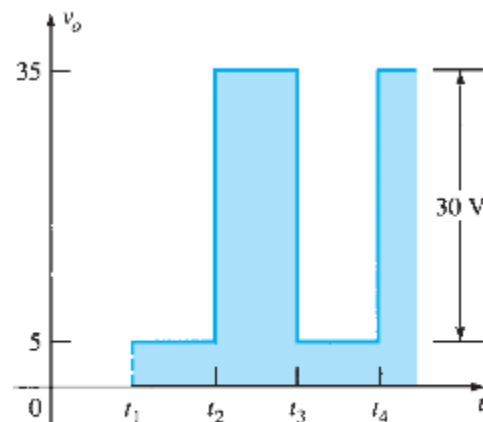
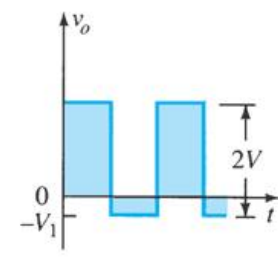
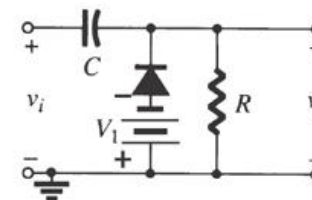
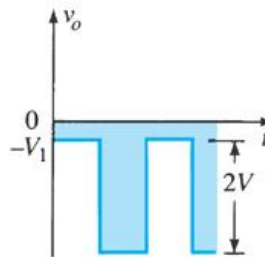
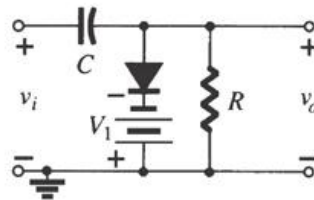
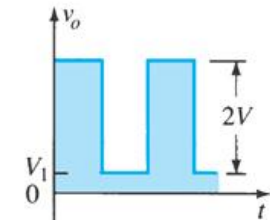
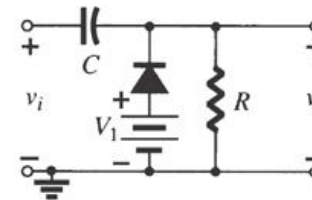
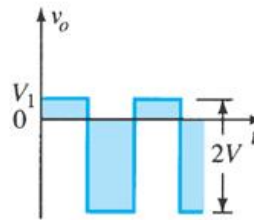
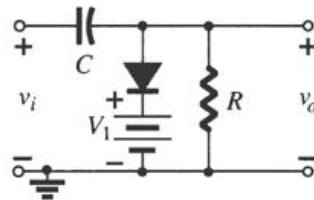
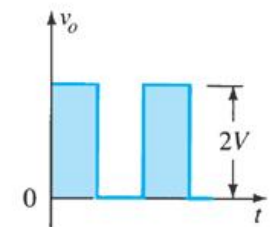
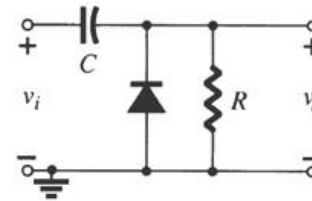
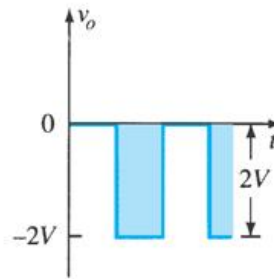
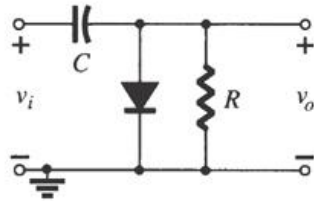
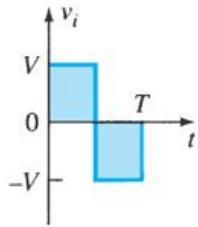


FIG. 2.95

Determining v_o with the diode in the “off” state.

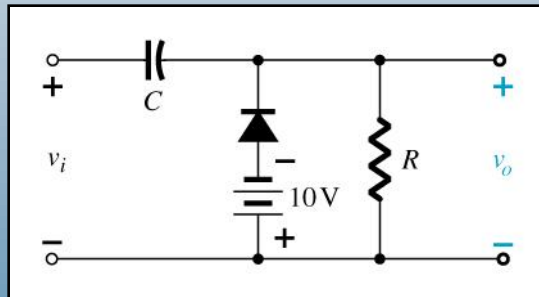
Summary of Clamper Circuits

Clamping Networks

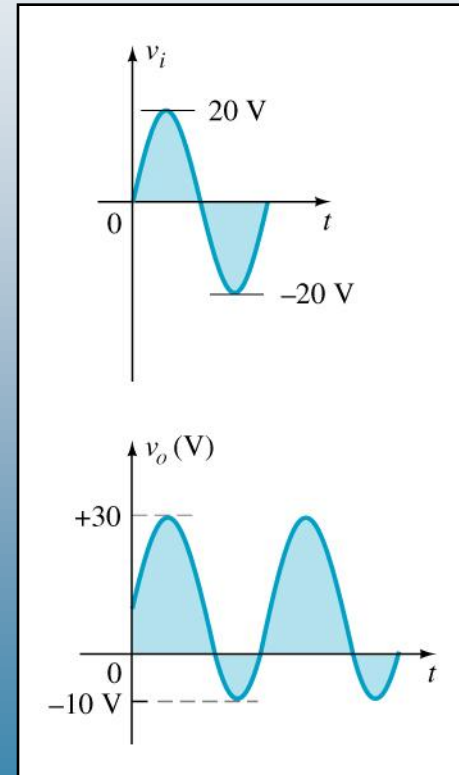


Biased Clamper Circuits with sine input

The input signal can be any type of waveform such as a sine, square, or triangle wave.



The DC source lets you adjust the DC clamping level.



2.11 Zener Diodes

Zener diode works exactly similar as Semiconductor diode in Forward Bias (FB)

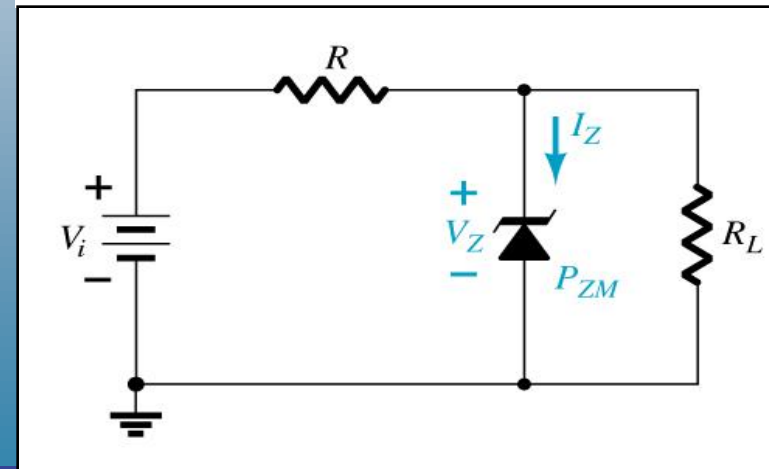
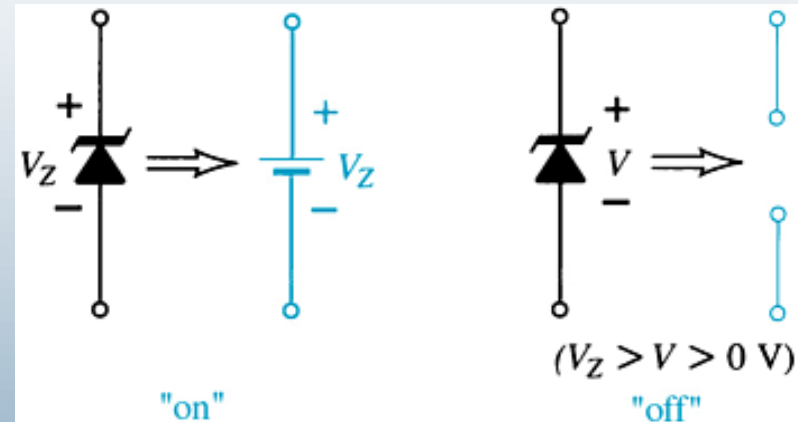
The Zener is a diode that is operated in reverse bias at the Zener Voltage (V_Z).

RB: When $V_i \geq V_Z$

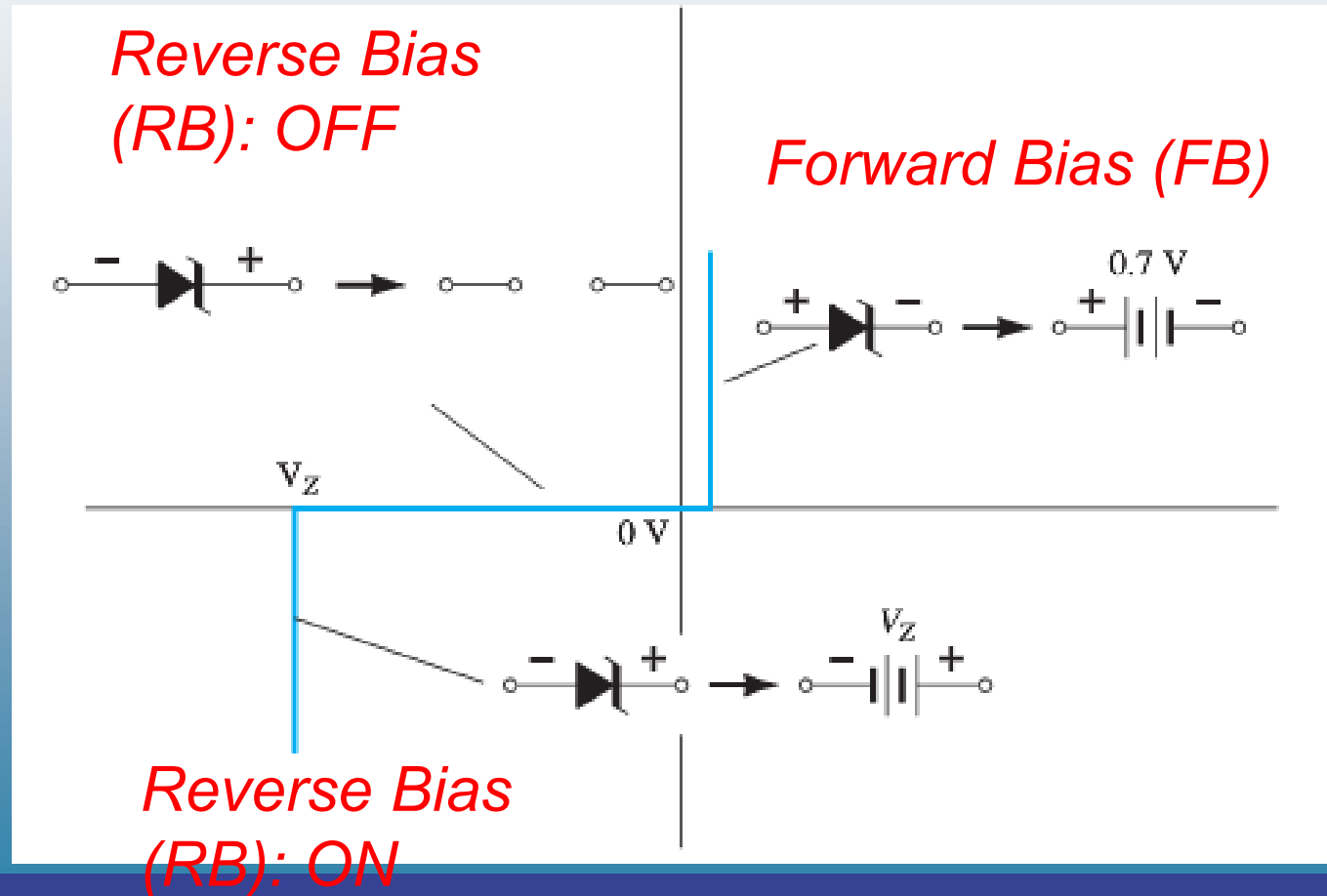
- The Zener is on
- Voltage across the Zener is V_Z
- Zener current: $I_Z = I_R - I_{RL}$
- The Zener Power: $P_Z = V_Z I_Z$

RB: When $V_i < V_Z$

- The Zener is off
- The Zener acts as an open circuit



Approximate equivalent circuits for the Zener diode in the three possible regions of application



EXAMPLE 2.24 Determine the reference voltages provided by the network of Fig. 2.109, which uses a white LED to indicate that the power is on. What is the level of current through the LED and the power delivered by the supply? How does the power absorbed by the LED compare to that of the 6-V Zener diode?

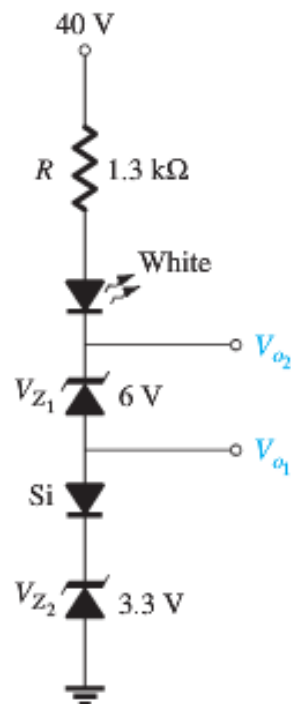


FIG. 2.109

Reference setting circuit for Example 2.24.

Note that the silicon diode was used to create a reference voltage of 4 V because

$$V_{o1} = V_{Z2} + V_K = 3.3 \text{ V} + 0.7 \text{ V} = \mathbf{4.0 \text{ V}}$$

Combining the voltage of the 6-V Zener diode with the 4 V results in

$$V_{o2} = V_{o1} + V_{Z1} = 4 \text{ V} + 6 \text{ V} = \mathbf{10 \text{ V}}$$

Finally, the 4 V across the white LED will leave a voltage of $40 \text{ V} - 14 \text{ V} = 26 \text{ V}$ across the resistor, and

$$I_R = I_{\text{LED}} = \frac{V_R}{R} = \frac{40 \text{ V} - V_{o2} - V_{\text{LED}}}{1.3 \text{ k}\Omega} = \frac{40 \text{ V} - 10 \text{ V} - 4 \text{ V}}{1.3 \text{ k}\Omega} = \frac{26 \text{ V}}{1.3 \text{ k}\Omega} = \mathbf{20 \text{ mA}}$$

which should establish the proper brightness for the LED.

The power delivered by the supply is simply the product of the supply voltage and current drain as follows:

$$P_s = EI_s = EI_R = (40 \text{ V})(20 \text{ mA}) = \mathbf{800 \text{ mW}}$$

The power absorbed by the LED is

$$P_{\text{LED}} = V_{\text{LED}} I_{\text{LED}} = (4 \text{ V})(20 \text{ mA}) = \mathbf{80 \text{ mW}}$$

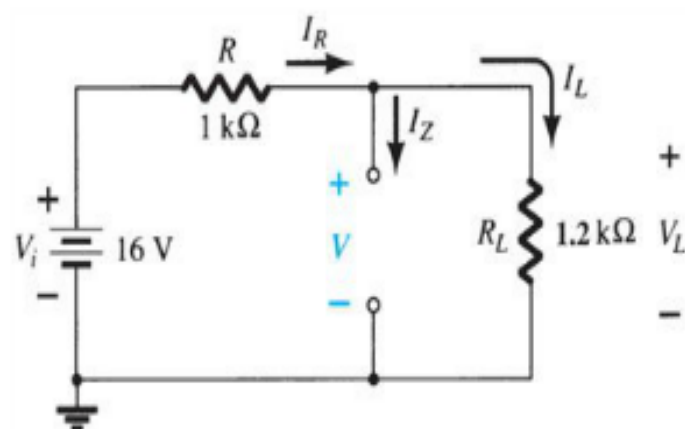
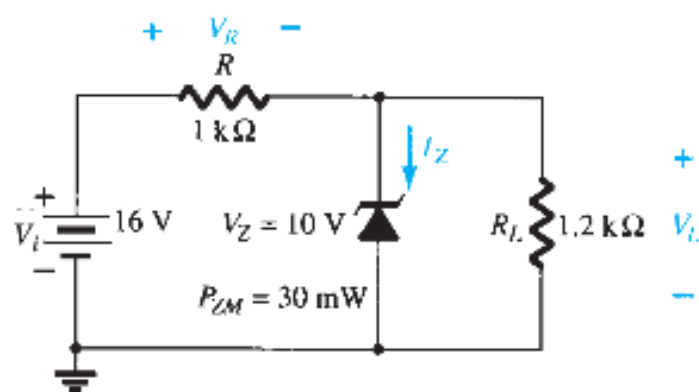
and the power absorbed by the 6-V Zener diode is

$$P_Z = V_Z I_Z = (6 \text{ V})(20 \text{ mA}) = \mathbf{120 \text{ mW}}$$

The power absorbed by the Zener diode exceeds that of the LED by 40 mW.

EXAMPLE 2.26

- a. For the Zener diode network of Fig. 2.115, determine V_L , V_R , I_Z , and P_Z .
 b. Repeat part (a) with $R_L = 3 \text{ k}\Omega$.



$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$

$$V_L = V = 8.73 \text{ V}$$

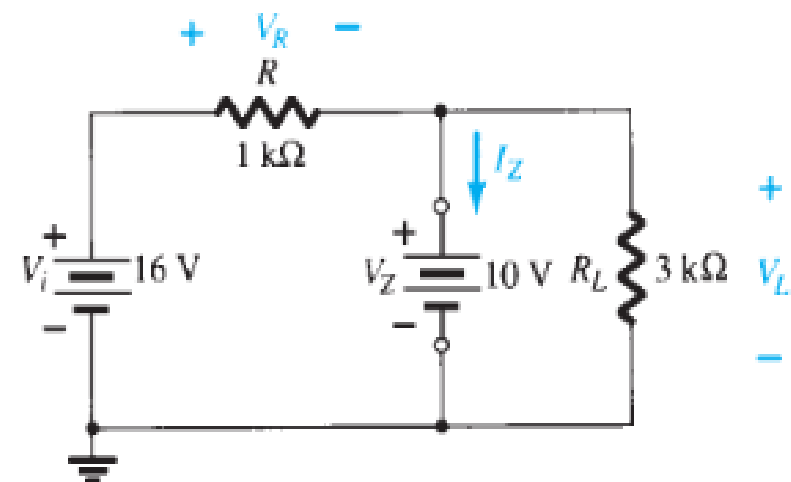
$$V_R = V_i - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

$$I_Z = 0 \text{ A}$$

$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$

EXAMPLE 2.26

- a. For the Zener diode network of Fig. 2.115, determine V_L , V_R , I_Z , and P_Z .
 b. Repeat part (a) with $R_L = 3 \text{ k}\Omega$.



$$V = \frac{R_L V_i}{R + R_L} = \frac{3 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 3 \text{ k}\Omega} = 12 \text{ V}$$

$$V_L = V_Z = \mathbf{10 \text{ V}}$$

$$V_R = V_i - V_L = 16 \text{ V} - 10 \text{ V} = \mathbf{6 \text{ V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{10 \text{ V}}{3 \text{ k}\Omega} = 3.33 \text{ mA}$$

$$I_R = \frac{V_R}{R} = \frac{6 \text{ V}}{1 \text{ k}\Omega} = 6 \text{ mA}$$

$$\begin{aligned} I_Z &= I_R - I_L \text{ [Eq. (2.18)]} \\ &= 6 \text{ mA} - 3.33 \text{ mA} \\ &= \mathbf{2.67 \text{ mA}} \end{aligned}$$

The power dissipated is

$$P_Z = V_Z I_Z = (10 \text{ V})(2.67 \text{ mA}) = \mathbf{26.7 \text{ mW}}$$

which is less than the specified $P_{ZM} = 30 \text{ mW}$.

EXAMPLE 2.27

- For the network of Fig. 2.119, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.
- Determine the maximum wattage rating of the diode.

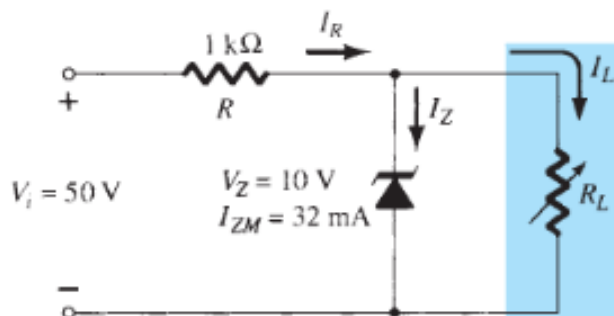


FIG. 2.119

Voltage regulator for Example 2.27.

Solution:

- To determine the value of R_L that will turn the Zener diode on, apply Eq. (2.2

$$R_{L_{\min}} = \frac{RV_Z}{V_i - V_Z} = \frac{(1\text{ k}\Omega)(10\text{ V})}{50\text{ V} - 10\text{ V}} = \frac{10\text{ k}\Omega}{40} = \mathbf{250\ \Omega}$$

The voltage across the resistor R is then determined by Eq. (2.22):

$$V_R = V_i - V_Z = 50\text{ V} - 10\text{ V} = \mathbf{40\text{ V}}$$

and Eq. (2.23) provides the magnitude of I_R :

$$I_R = \frac{V_R}{R} = \frac{40\text{ V}}{1\text{ k}\Omega} = \mathbf{40\text{ mA}}$$

The minimum level of I_L is then determined by Eq. (2.25):

$$I_{L_{\min}} = I_R - I_{ZM} = 40\text{ mA} - 32\text{ mA} = \mathbf{8\text{ mA}}$$

with Eq. (2.26) determining the maximum value of R_L :

$$R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}} = \frac{10\text{ V}}{8\text{ mA}} = \mathbf{1.25\text{ k}\Omega}$$

$$\begin{aligned} \text{b. } P_{\max} &= V_Z I_{ZM} \\ &= (10\text{ V})(32\text{ mA}) = \mathbf{320\text{ mW}} \end{aligned}$$

EXAMPLE 2.28 Determine the range of values of V_i that will maintain the Zener diode of Fig. 2.121 in the “on” state.

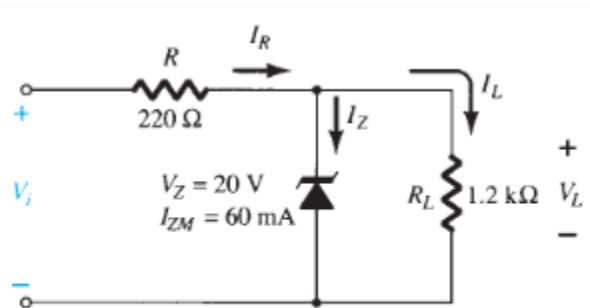


FIG. 2.121

Regulator for Example 2.28.

Solution:

$$\text{Eq. (2.27): } V_{i_{\min}} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200 \, \Omega + 220 \, \Omega)(20 \, \text{V})}{1200 \, \Omega} = \mathbf{23.67 \, \text{V}}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20 \, \text{V}}{1.2 \, \text{k}\Omega} = 16.67 \, \text{mA}$$

$$\begin{aligned} \text{Eq. (2.28): } I_{R_{\max}} &= I_{ZM} + I_L = 60 \, \text{mA} + 16.67 \, \text{mA} \\ &= 76.67 \, \text{mA} \end{aligned}$$

$$\begin{aligned} \text{Eq. (2.29): } V_{i_{\max}} &= I_{R_{\max}}R + V_Z \\ &= (76.67 \, \text{mA})(0.22 \, \text{k}\Omega) + 20 \, \text{V} \\ &= 16.87 \, \text{V} + 20 \, \text{V} \\ &= \mathbf{36.87 \, \text{V}} \end{aligned}$$

A plot of V_L versus V_i is provided in Fig. 2.122.

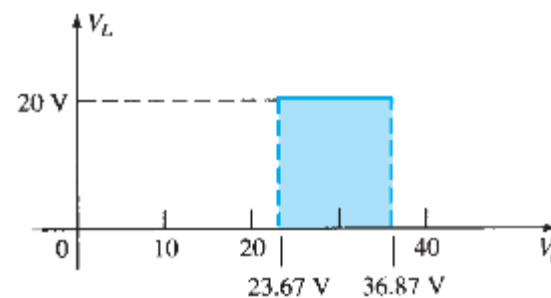


FIG. 2.122

V_L versus V_i for the regulator of Fig. 2.121.

- Please write the following things on your exam paper and take your CLEAR picture with the First Page.
- You can take the picture and save it on your phone NOW.

1. Write your Name and NSU ID.
2. Write the Course Name, Course Code and section number. Analog Electronics I (EEE111.8)
3. Write the following pledge. "I affirm that I have not given or received any unauthorized help on this exam, and that all work are my own".
4. Clearly sign below the pledge.
5. Give page numbers for ATLEAST 8 pages

- $X = 6^{\text{th}}$ digit of your NSU ID
- Example: If 1632345042 is your ID, then $X = 4$.
- You MUST have to submit before 12:50 PM
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- If you miss any page, send me a private comment ASAP